

Transforming Diffusion

How transformer models are revolutionizing image synthesis

Thomas Doods, Alexander Belousov

Context

Research project 2 (12 sp)

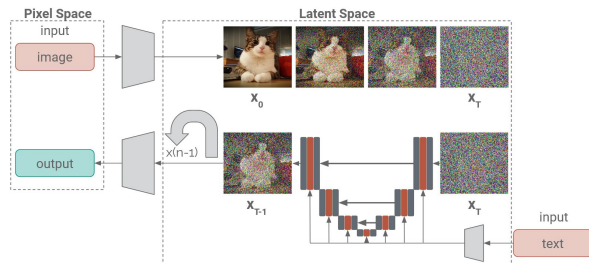
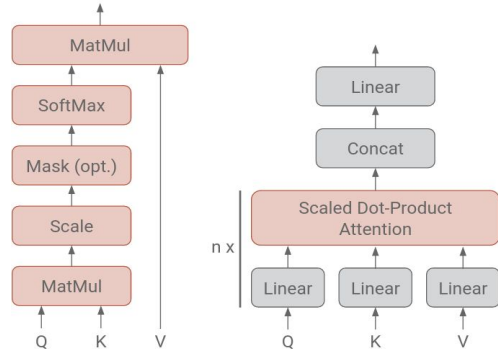
- **Transformers United**
- Deep multi-task & meta-learning



Content

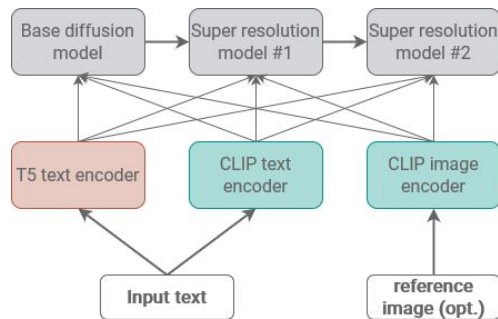
Are you wasting your time or will it actually be interesting?

The Basics



Diffusion Models

Extensions



Attention

[insert somewhat funny joke about paying attention]

Introducing transformers

Attention is all you need
Vaswani, Ashish et al. (2017)

Illustrated: Self-Attention
Raimi Karim (2019)



Input $\Rightarrow$ key, value and query

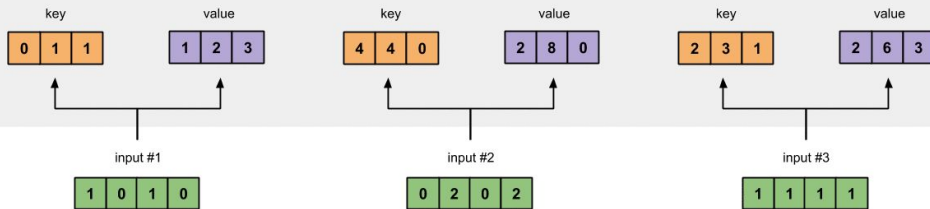
$$\begin{bmatrix} \overrightarrow{\text{input}_1} \\ \overrightarrow{\text{input}_2} \\ \overrightarrow{\text{input}_3} \end{bmatrix} \times W_k = \begin{bmatrix} \overrightarrow{\text{key}_1} \\ \overrightarrow{\text{key}_2} \\ \overrightarrow{\text{key}_3} \end{bmatrix} \quad \begin{bmatrix} \overrightarrow{\text{input}_1} \\ \overrightarrow{\text{input}_2} \\ \overrightarrow{\text{input}_3} \end{bmatrix} \times W_v = \begin{bmatrix} \overrightarrow{\text{value}_1} \\ \overrightarrow{\text{value}_2} \\ \overrightarrow{\text{value}_3} \end{bmatrix} \quad \begin{bmatrix} \overrightarrow{\text{input}_1} \\ \overrightarrow{\text{input}_2} \\ \overrightarrow{\text{input}_3} \end{bmatrix} \times W_q = \begin{bmatrix} \overrightarrow{\text{query}_1} \\ \overrightarrow{\text{query}_2} \\ \overrightarrow{\text{query}_3} \end{bmatrix}$$

Attention scores = query $\times$ keys

Self-attention

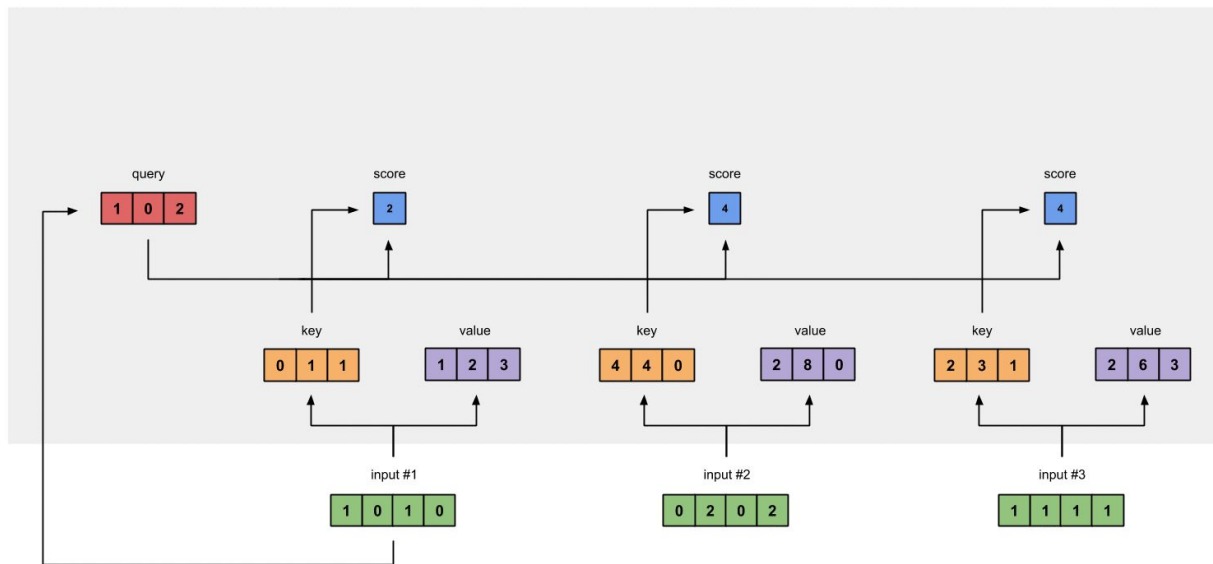
$$\begin{bmatrix} \overrightarrow{query_1} \\ \overrightarrow{query_2} \\ \overrightarrow{query_3} \end{bmatrix} \times \begin{bmatrix} \overrightarrow{key_1} \\ \overrightarrow{key_2} \\ \overrightarrow{key_3} \end{bmatrix}^T = \begin{bmatrix} \overrightarrow{score_1} \\ \overrightarrow{score_2} \\ \overrightarrow{score_3} \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 2 \\ \overrightarrow{query_2} \\ \overrightarrow{query_3} \end{bmatrix} \times \begin{bmatrix} 0 & 4 & 2 \\ 1 & 4 & 3 \\ 1 & 0 & 1 \end{bmatrix}^T = \begin{bmatrix} 2 & 4 & 4 \\ \overrightarrow{score_2} \\ \overrightarrow{score_3} \end{bmatrix}$$



Softmax scores

Self-attention



Multiply with values

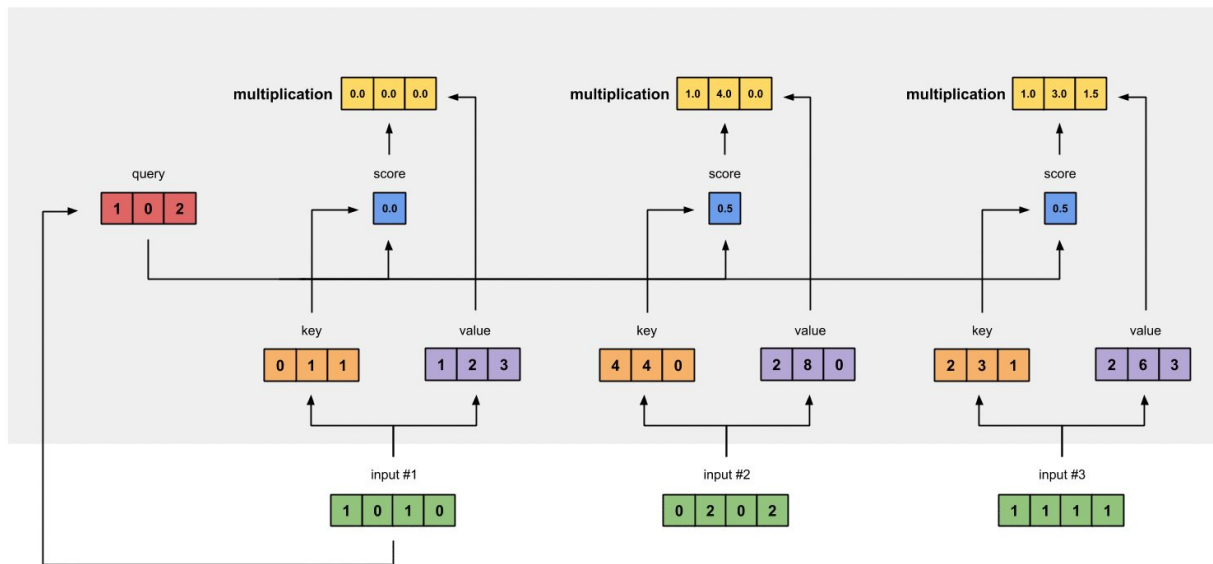
$$0.0 \times \begin{bmatrix} 1 & 2 & 3 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \end{bmatrix}$$

$$0.5 \times \begin{bmatrix} 2 & 8 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 4 & 0 \end{bmatrix}$$

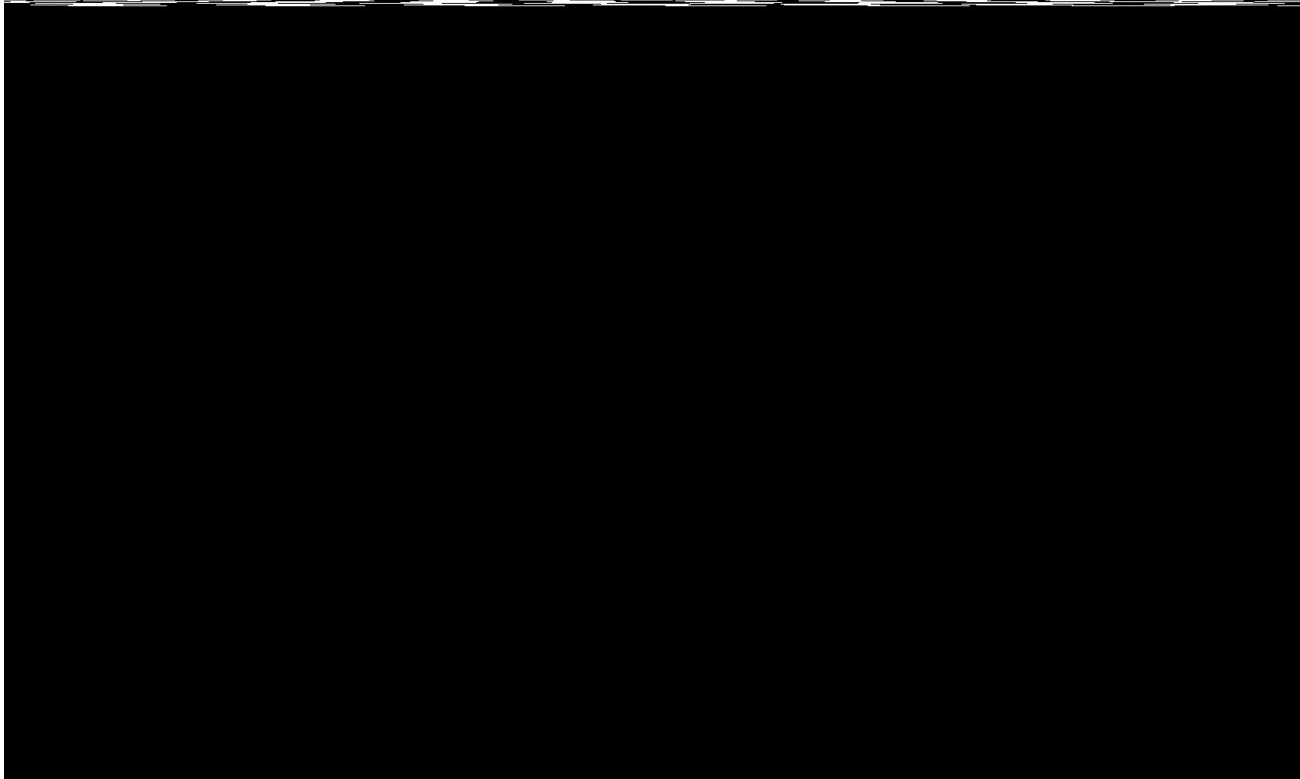
$$0.5 \times \begin{bmatrix} 2 & 6 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 3 & 1.5 \end{bmatrix}$$

Sum score weighted values into output

Self-attention



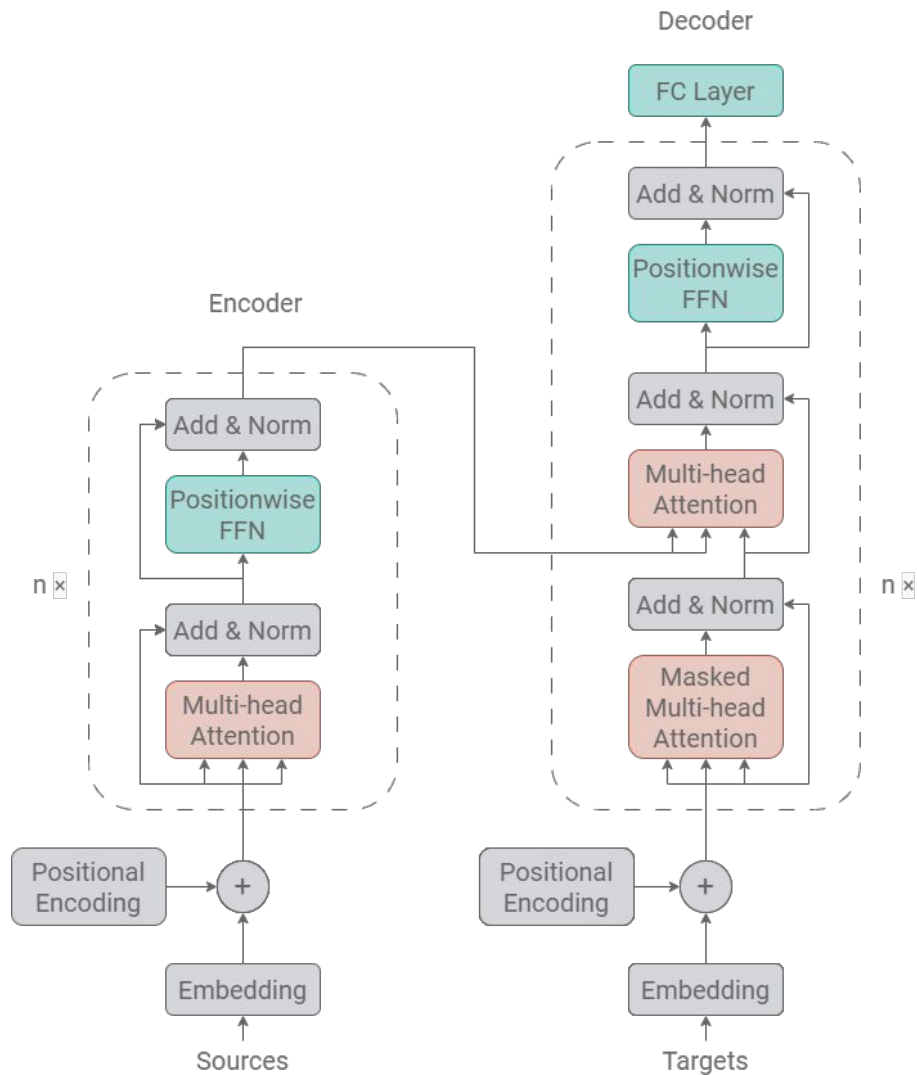
Repeat for other inputs



Transformers

What is all the hype about?

Architecture



Embeddings

"I have a cat."

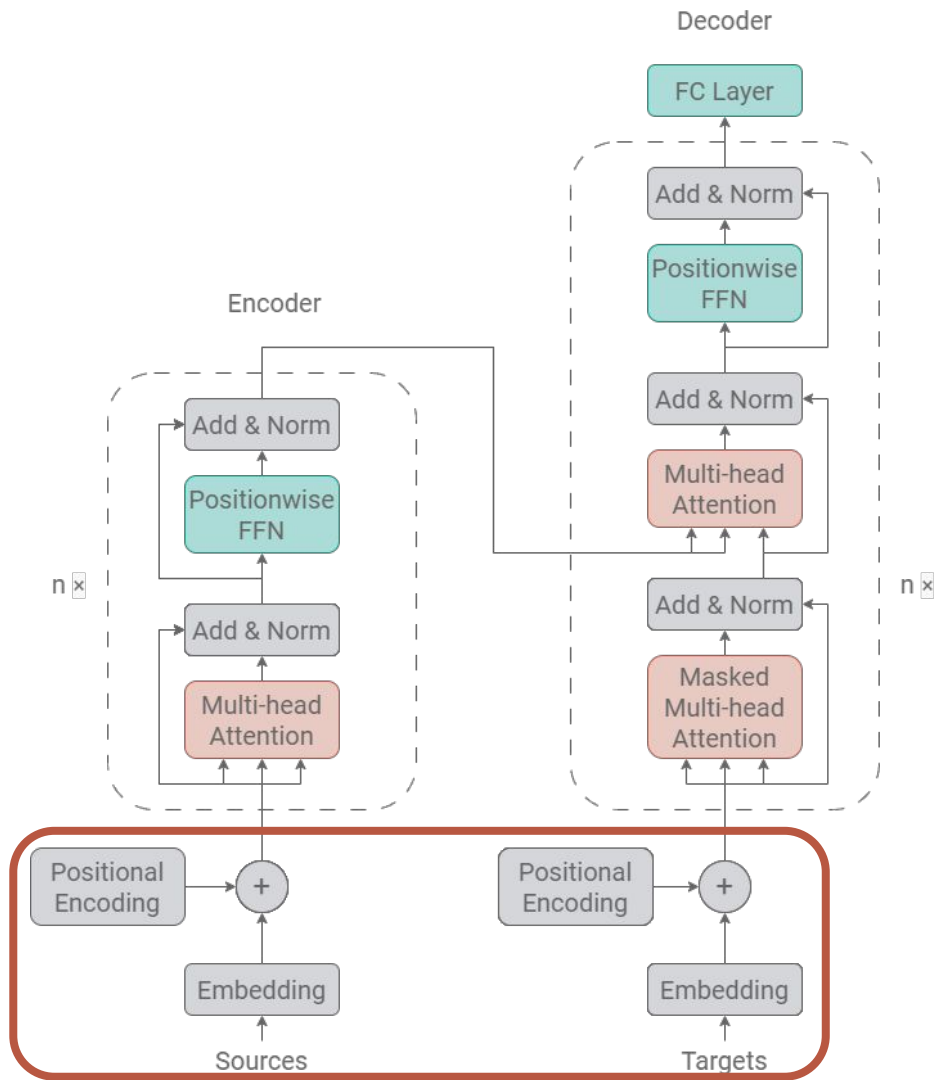
Tokenizer

⇒ "i", "have", "a", "cat", "."

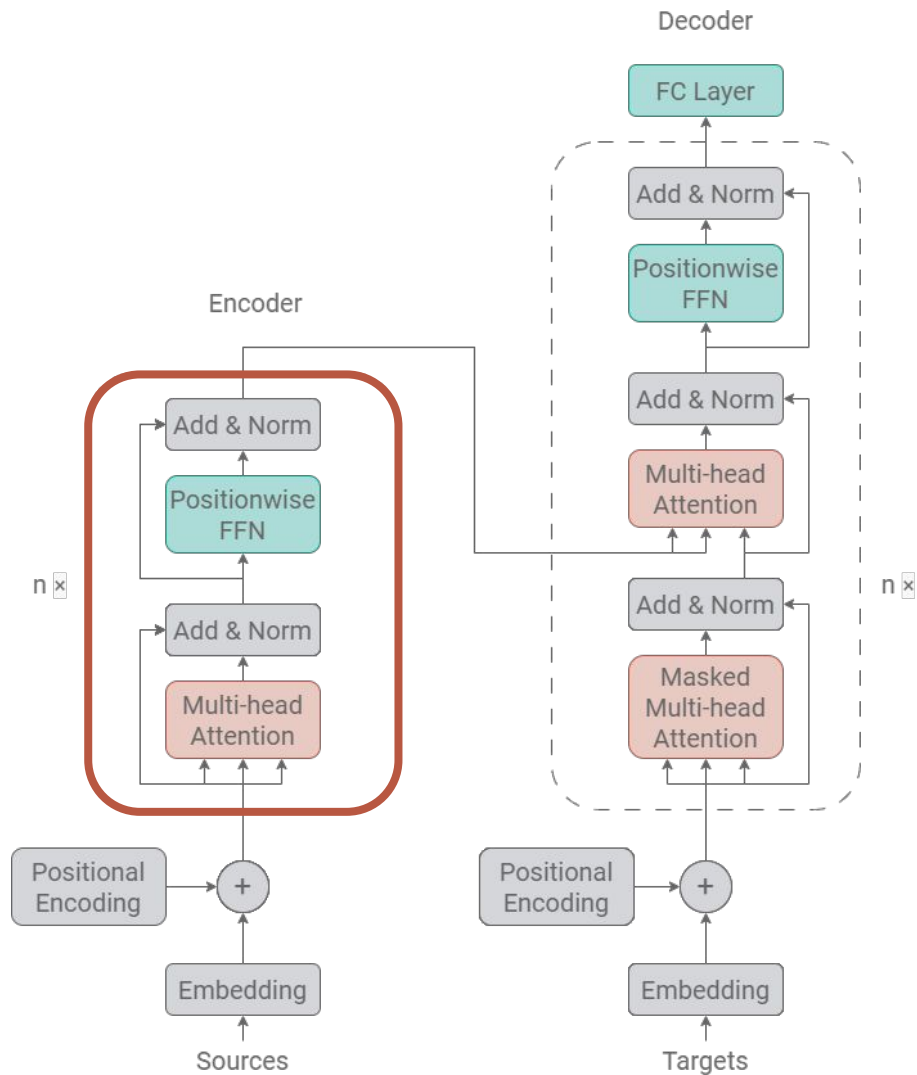
⇒ 101, 432, 67, 43, 407

Linear

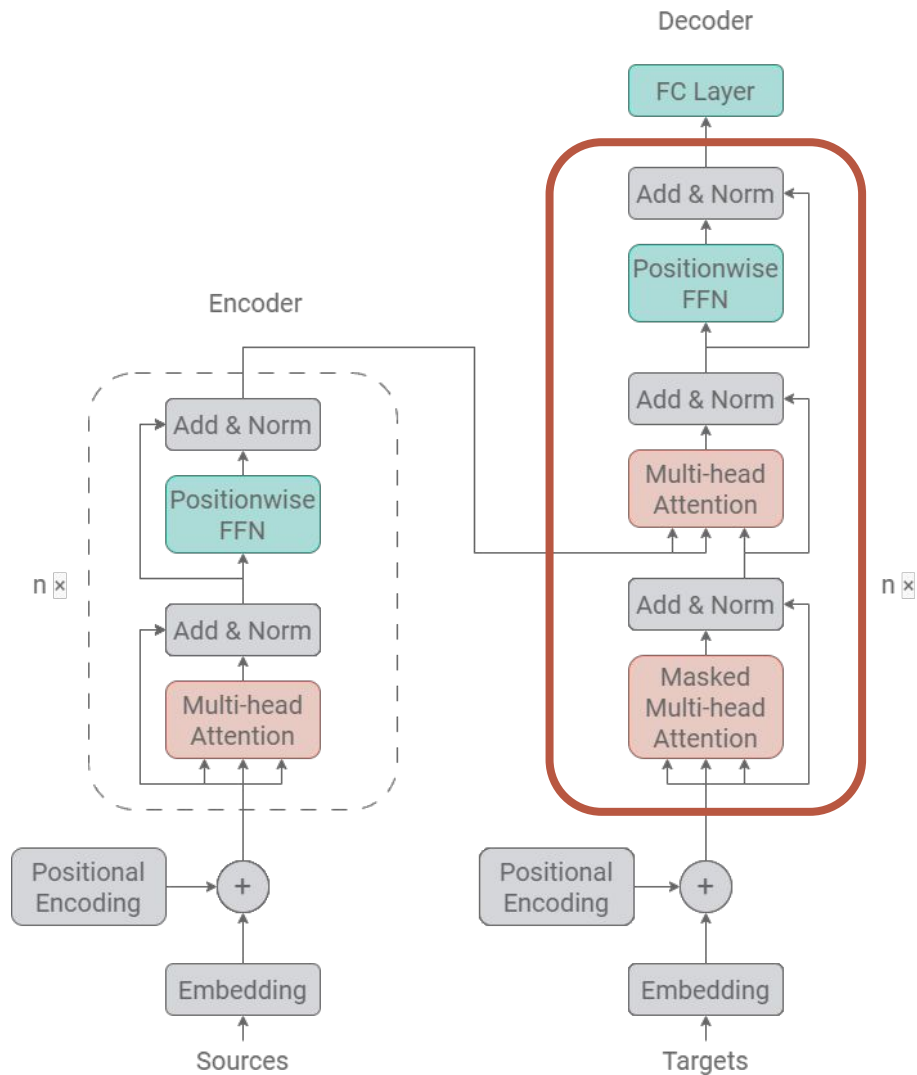
⇒ $E_{101}, E_{432}, E_{67}, E_{43}, E_{407}$



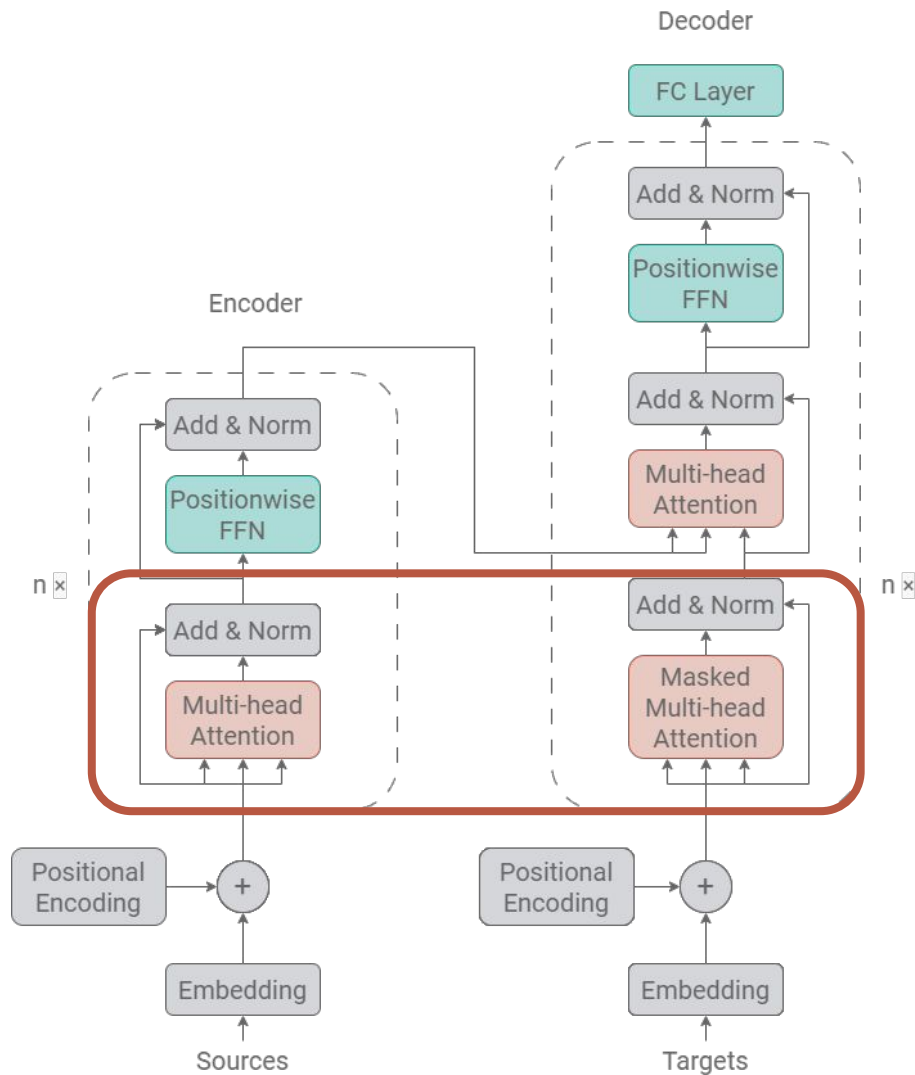
Encoder block



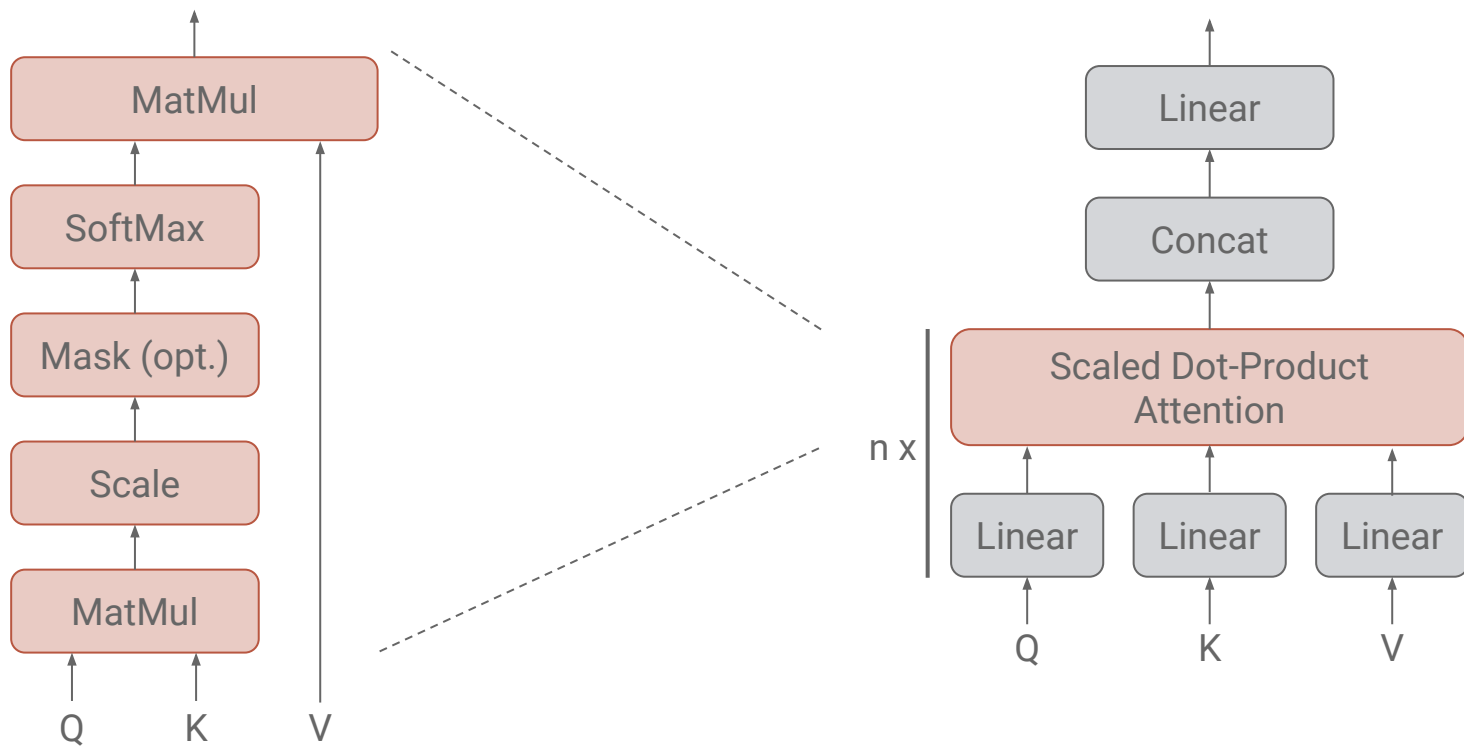
Decoder block



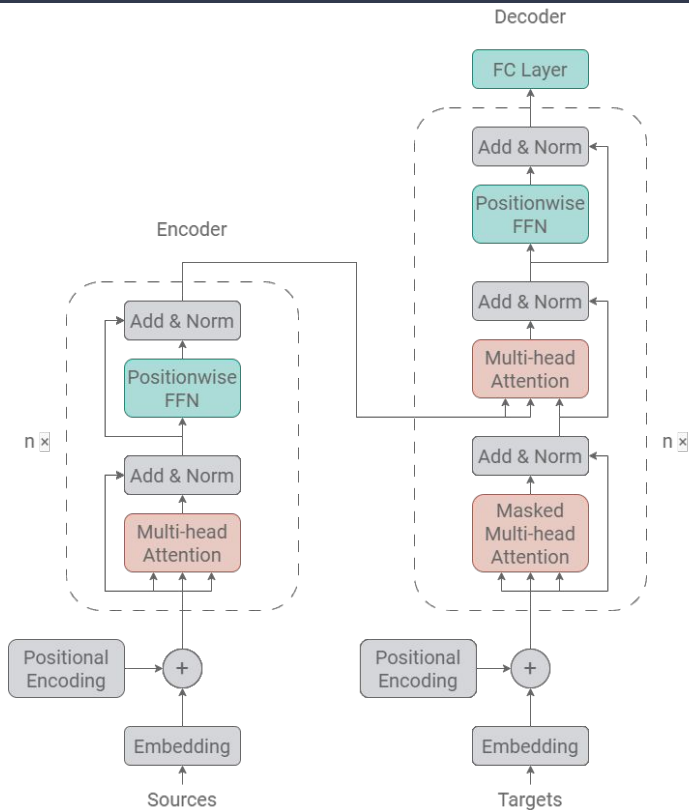
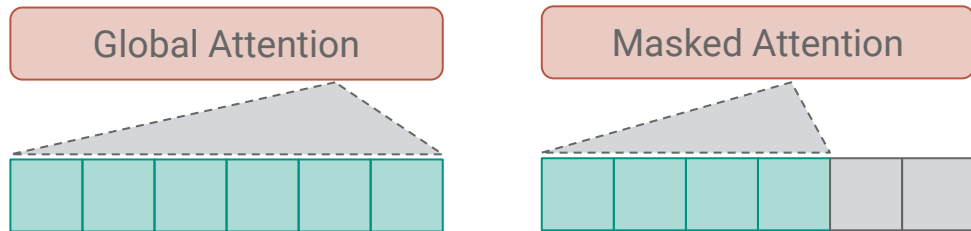
Self Attention



Multi-Head Attention



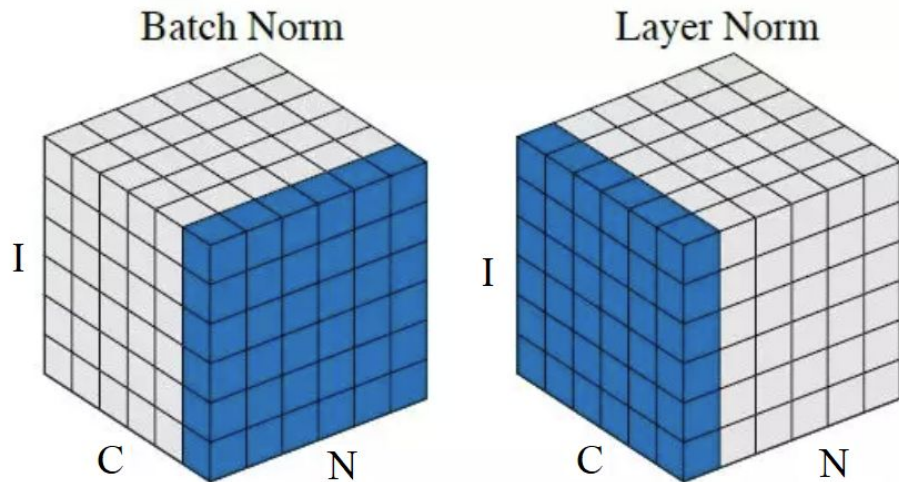
Masked Self-Attention



Layernorm

No more dependence on batches and
moving averages

Layer Normalization
Ba, Kiros, Hinton (2016)



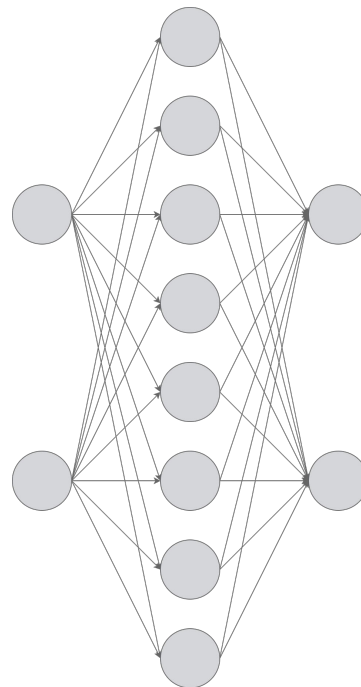
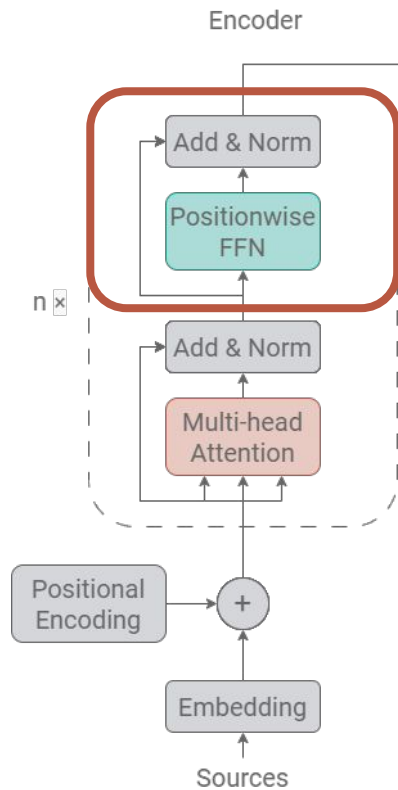
$$H = I \times C$$

$$\mu = \frac{1}{H} \sum_{i=1}^H x_i$$

$$\sigma^2 = \frac{1}{H} \sum_{i=1}^H (x_i - \mu)^2$$

$$\text{layernorm}(x_i) = \frac{x_i - \mu}{\sigma + \epsilon}$$

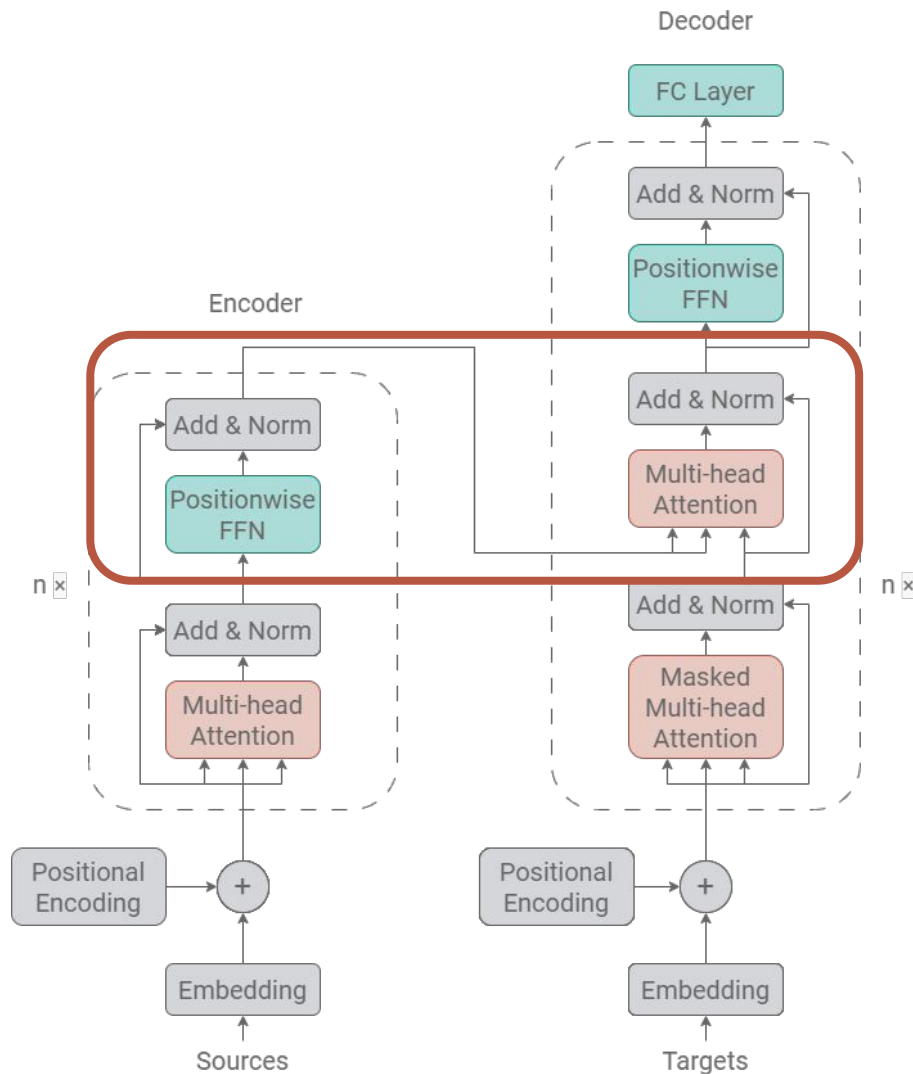
Positionwise Feed-Forward Network



Cross Attention

Queries from same block

Keys & values from external source

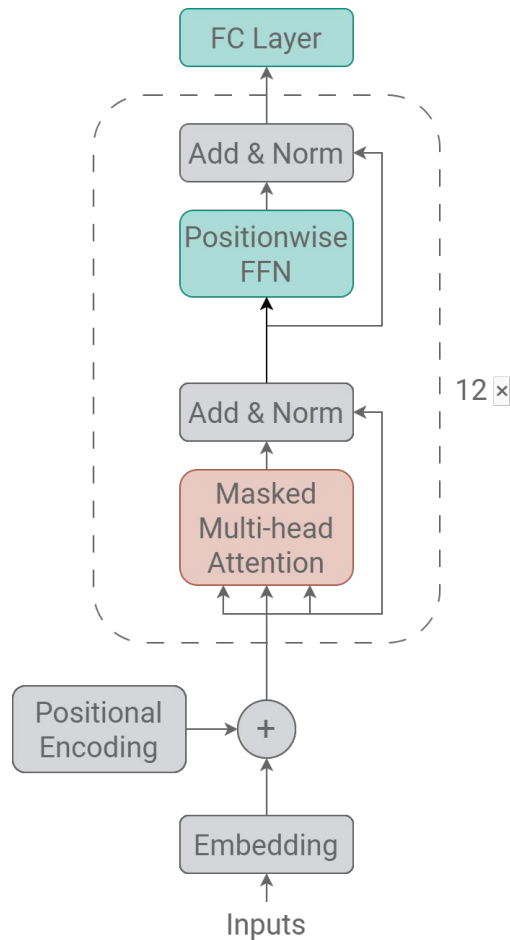


Encoders and Decoders in practice

Encoders are good at understanding context

Decoders are good at predicting or generating

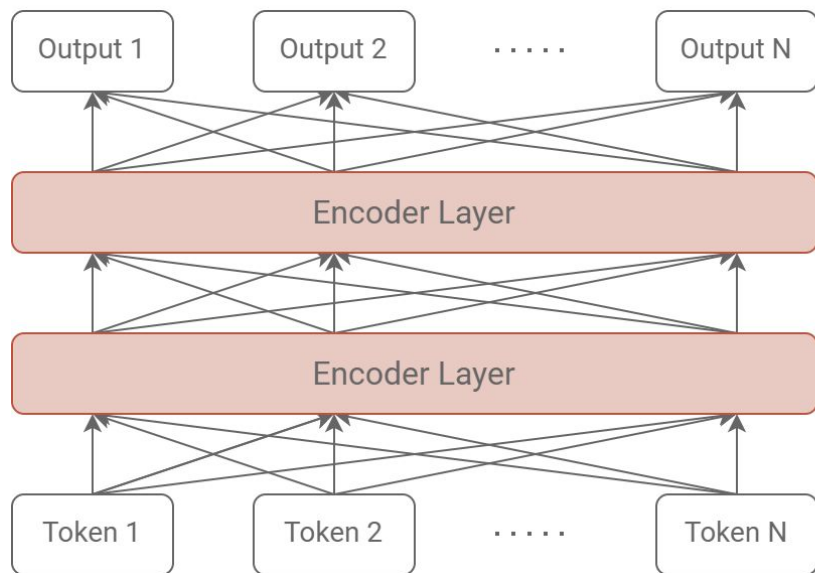
Encoder-decoders (seq to seq) are good at translation/summarization



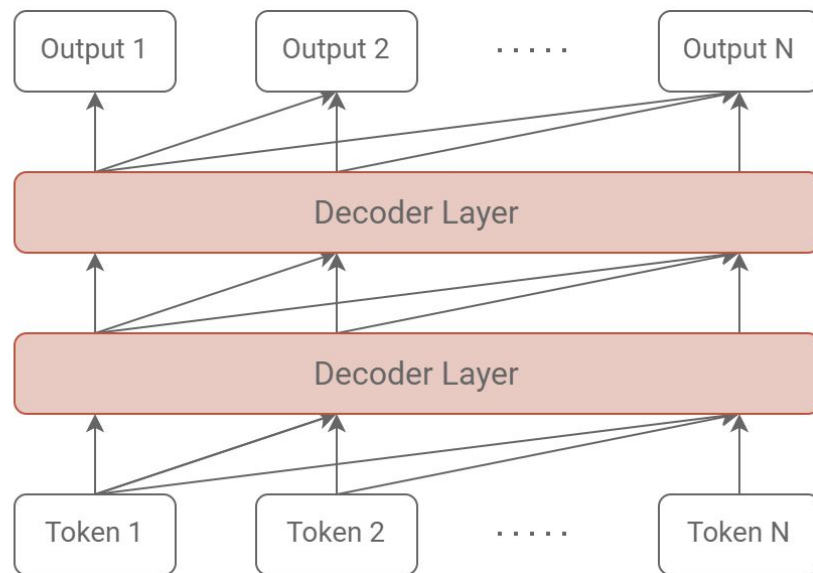
GPT architecture

Bidirectional vs unidirectional

BERT



GPT

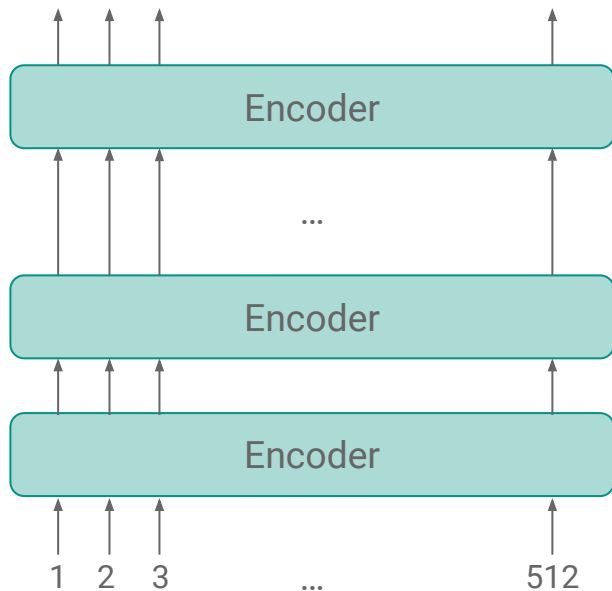


Encoders and Decoders in practice

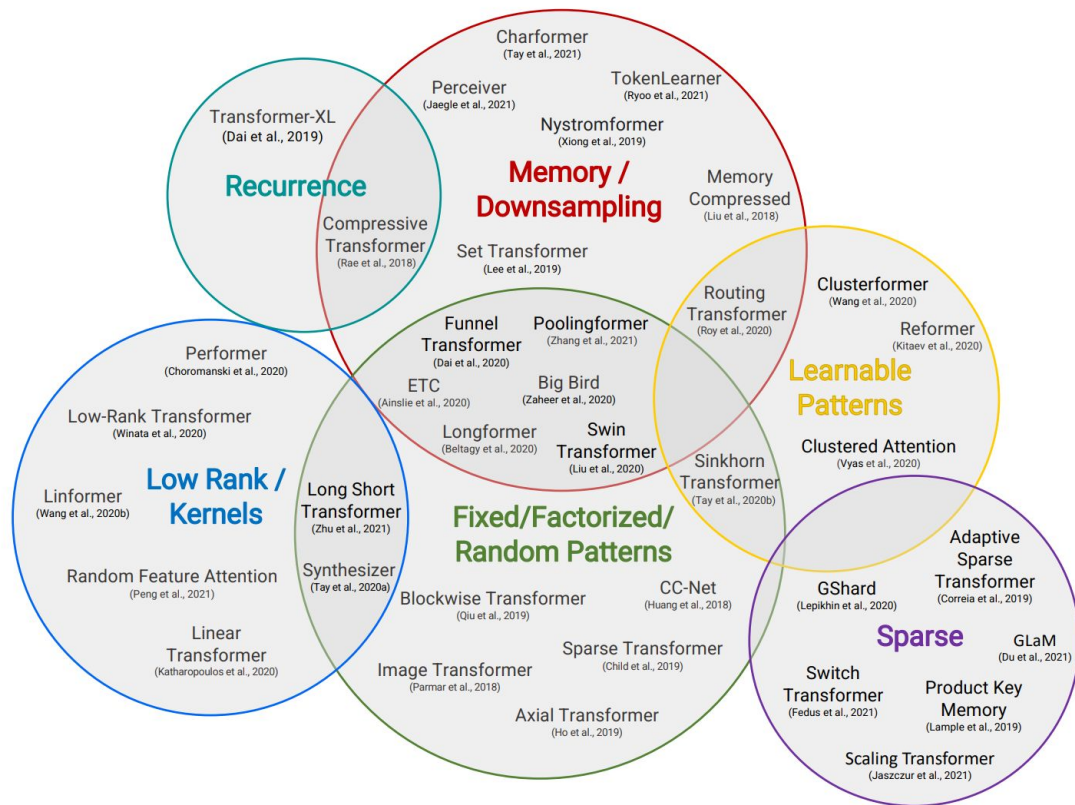
Encoders are good at understanding context

Decoders are good at predicting or generating

Encoder-decoders (seq to seq) are good at translation/summarization



Many different flavours



Recap

What did we see thus far?

Attention Principles

3 matrices learnt for transformations into

- Query
- Key
- Value

Self-Attention vs Cross-Attention

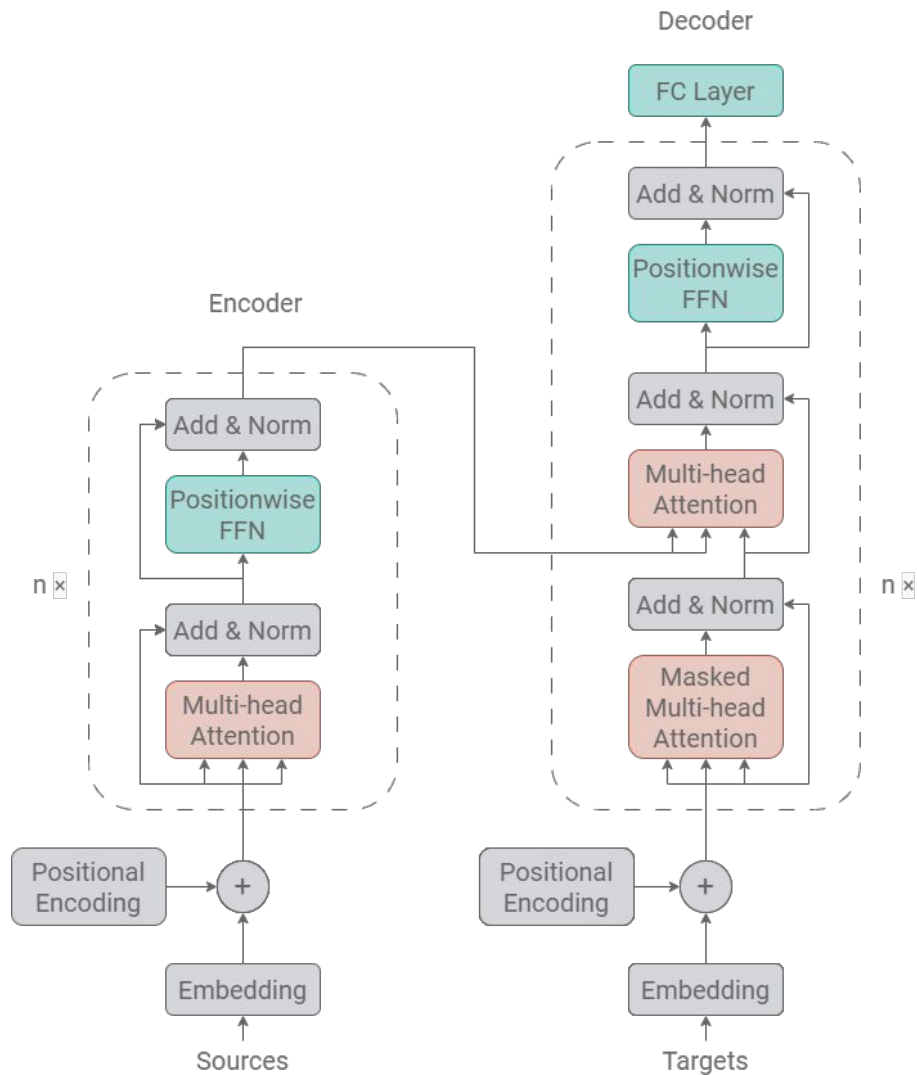
- Cross-Attention in decoder only
- Key and Value from encoder

Global Attention vs Masked Attention

Multi-head Attention

- Repeat and concatenate

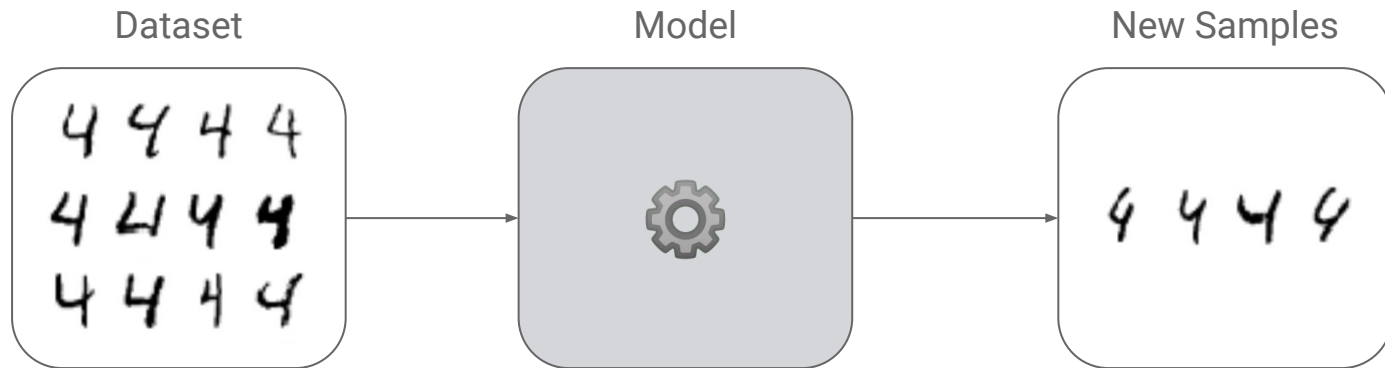
Transformer Architecture



Diffusion

How did image synthesis evolve during the past years?

Task: Generating images



Task: Generating images

Alternate formulation

Sampling from a distribution

The space is all possible combinations of pixels

The distribution is all compositions perceived as '4'

Challenges

Complex distribution, even for MNIST ($2^{28 \times 28}$)

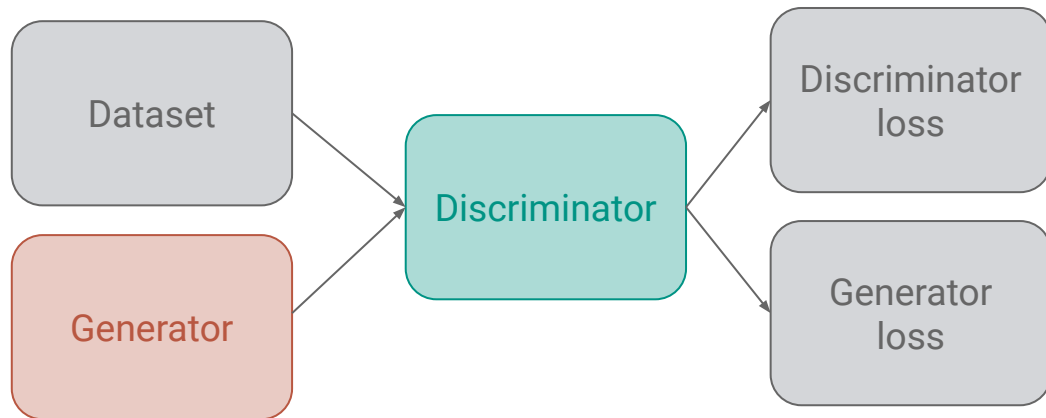
Issues

Simple distributions are not flexible enough to fit arbitrary data

Complex distributions are intractable and must resort to Monte Carlo processes

Generative adversarial networks

Generative Adversarial Networks
Ian Goodfellow et al. (2014)



Generated	Discriminator		Real
			
			
			

StyleGAN

A Style-Based Generator Architecture
for Generative Adversarial Networks
Karras et al. (2018)

thispersondoesnotexist.com



Large improvements to generator explainability
and accuracy

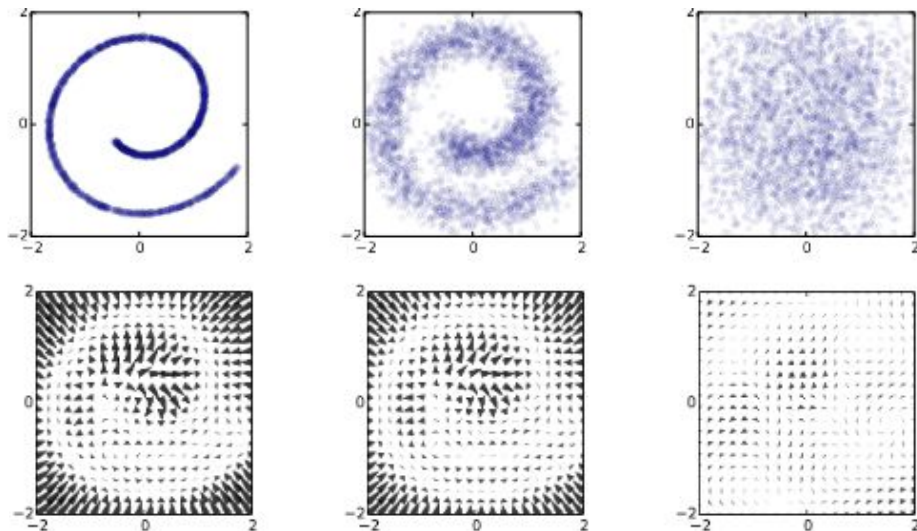
Probabilistic Diffusion Models

Deep Unsupervised Learning using
Nonequilibrium Thermodynamics
Sohl-Dickstein, Weiss et al. (2015)

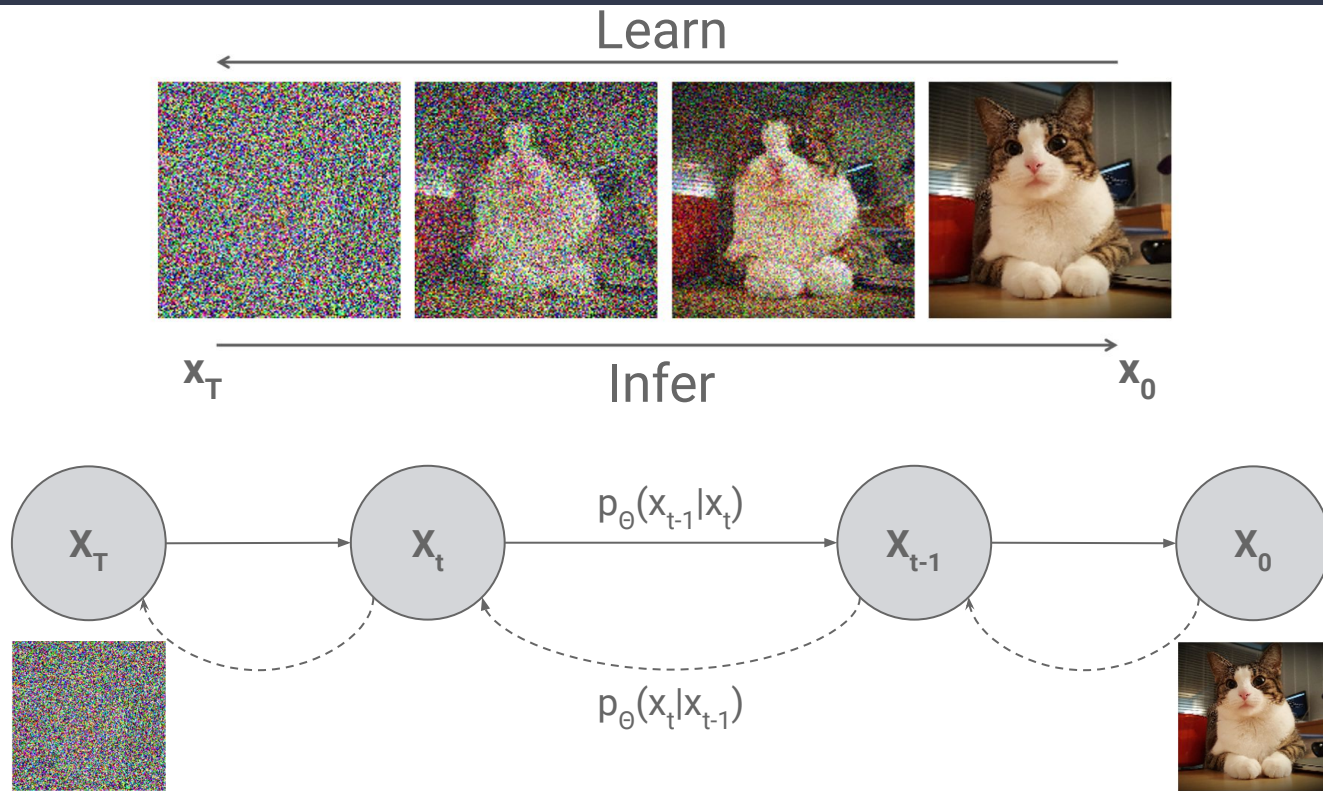
'Diffuse' a complex distribution to a simple one with
another simple distribution

Each forward step is analytically evaluable and the
same holds for the reverse step

Estimating small perturbations is easier than the
whole distribution at once



Diffusion & reverse process



Denoising DPM

Denoising Diffusion Probabilistic
Models

Ho, Jain, Abbeel (Dec 2020)

Denoising

Predicts the difference (noise) instead of the next step

Architecture

- U-net with self-attention (PixelCNN++)
- Step t is represented using sinusoidal embedding

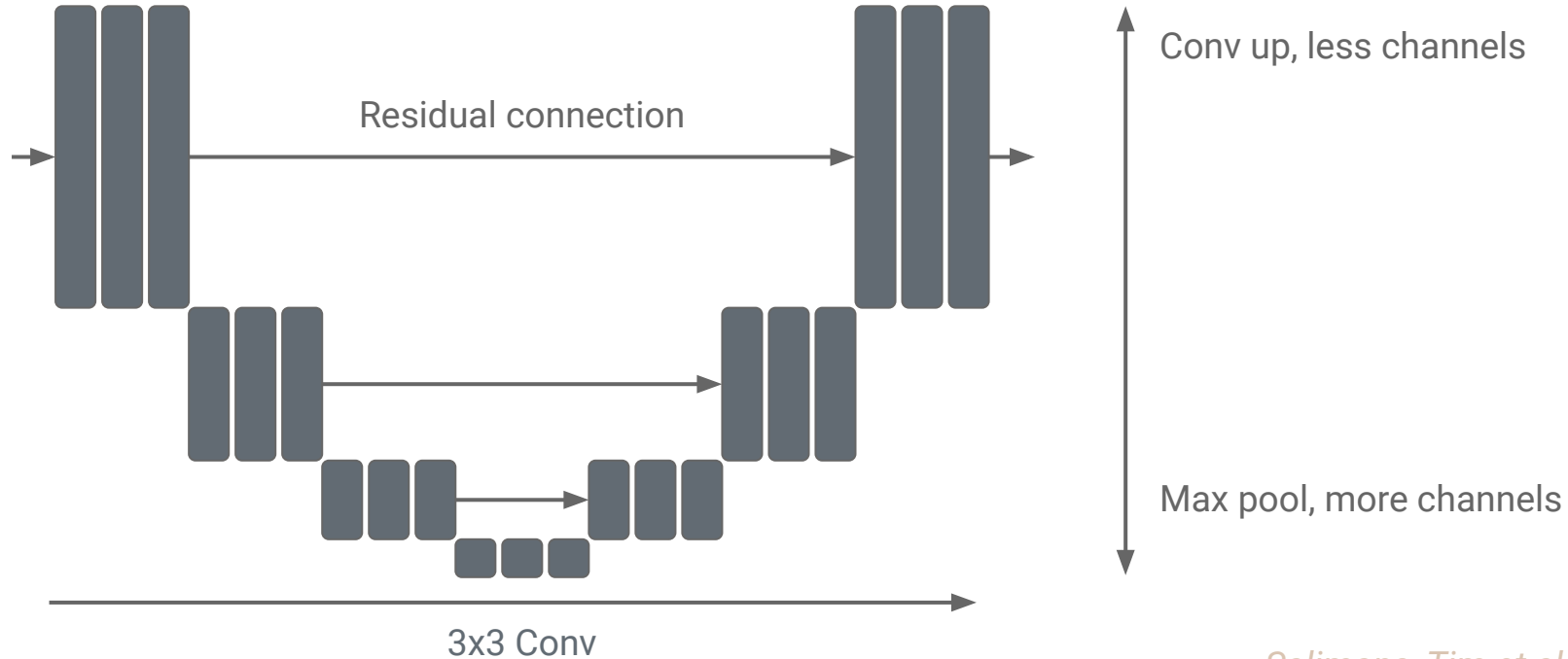
Details

- 500+ timesteps required
- Trained on CIFAR10

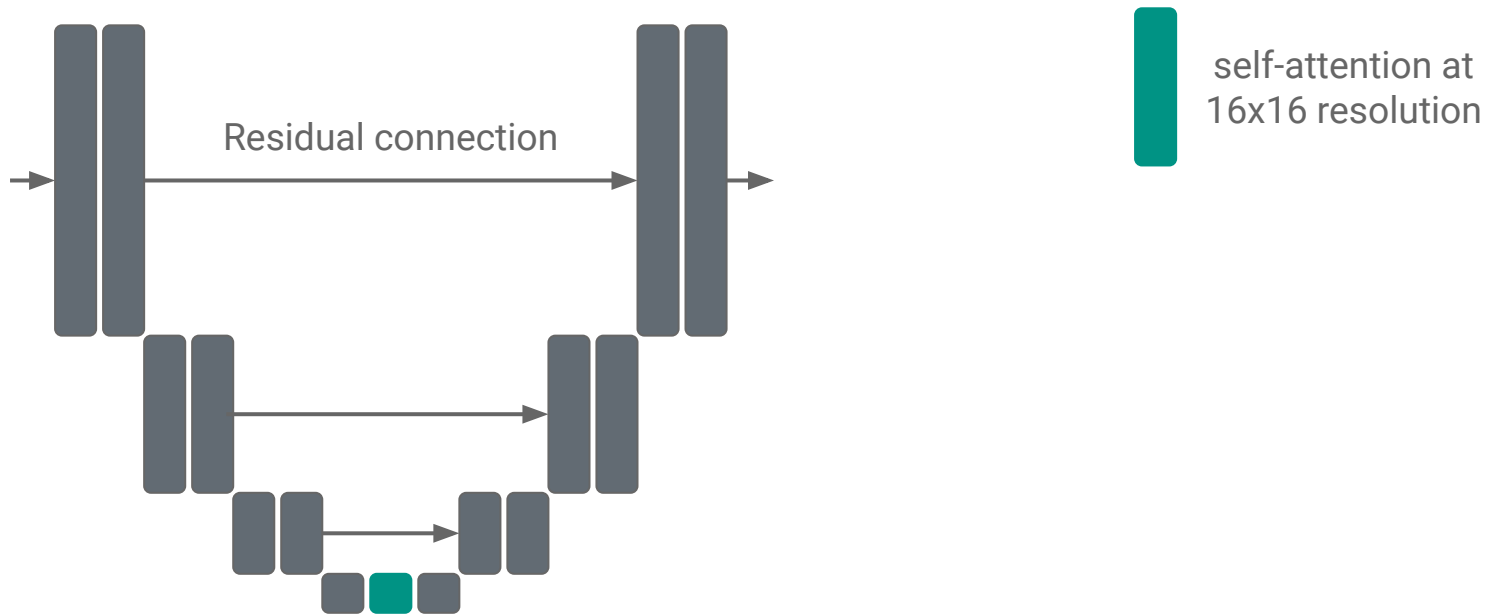
$$q(x_1, \dots, x_T | x_0) := \prod_{t=1}^T q(x_t | x_{t-1})$$

$$q(x_t | x_{t-1}) := \mathcal{N}(x_t; \sqrt{1 - \beta_t} x_{t-1}, \beta_t I)$$

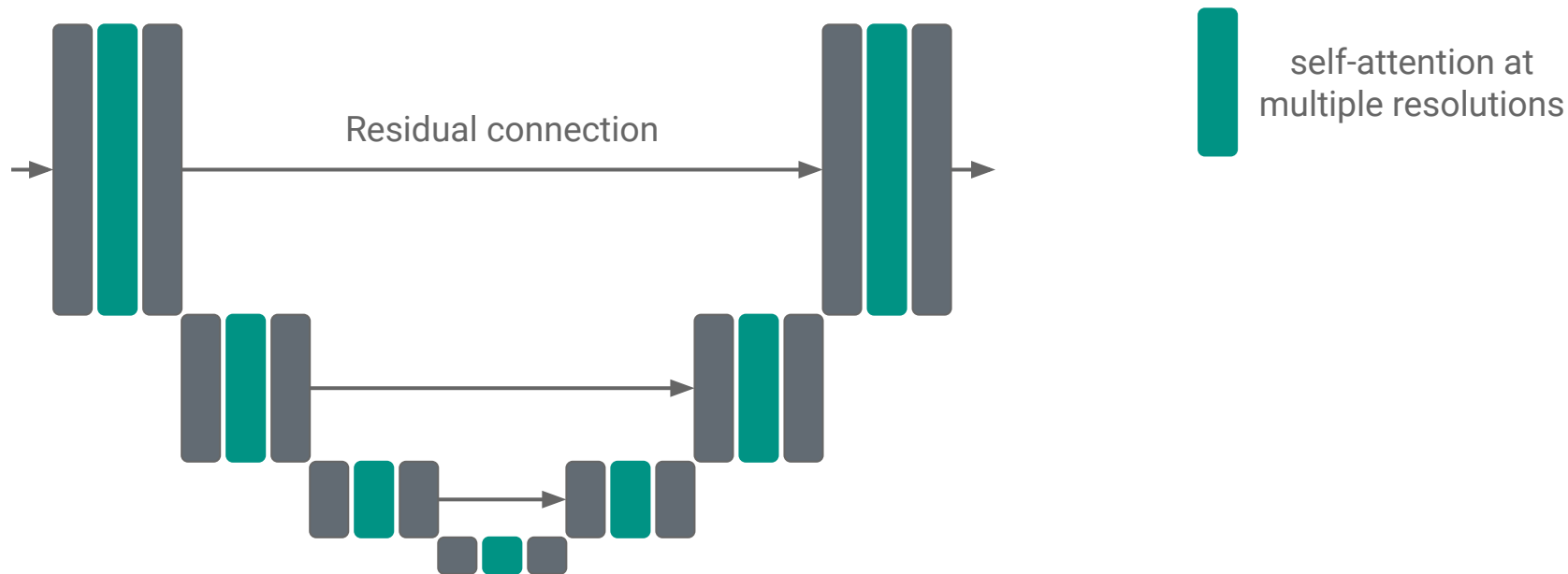
U-net



U-net with self attention (PixelCNN++)



U-net with self attention (PixelCNN++)

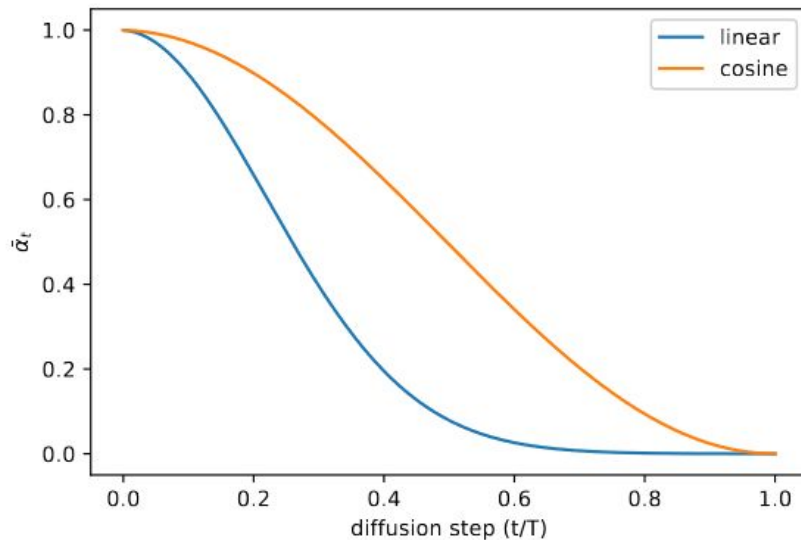


The rise of diffusion models

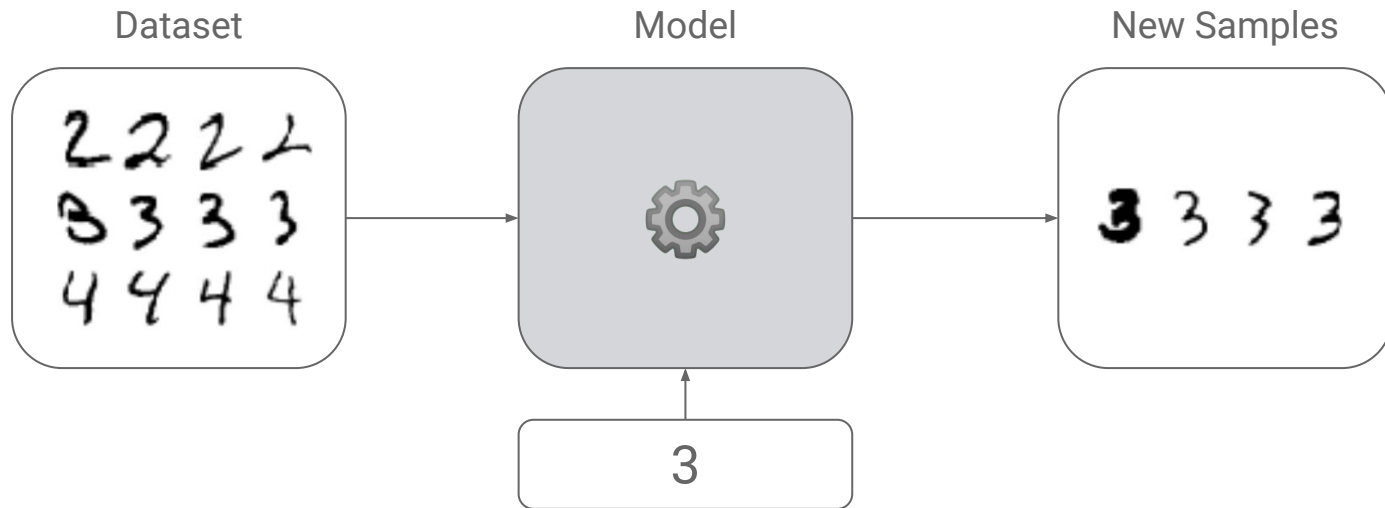
Improved Denoising Diffusion Probabilistic Models
Dhariwal, Nichol (Feb 2021)

Schedules

- Noise at the start is most important
- Destroy image slower at start to emphasise learning



Task: Conditional image synthesis



Diversity vs Inception Score

Diffusion Models Beat GANs on Image
Synthesis

Dhariwal, Nichol (Jun 2021)

Classifier-free diffusion guidance

Ho, Salimans (Jul 2022)*

**Presented at NeurIPS Dec 2021*

Classifier Guidance

- Analyze the image with an independent classifier
- Nudge each step toward its gradients
- Requires an extra trained classifier on noisy data

Classifier Free Guidance

- Calculate unconditional (null label) update step
- Nudge the step away from this prediction

Fidelity trade-off

- Using C(F)G creates better images
- But it has less diversity

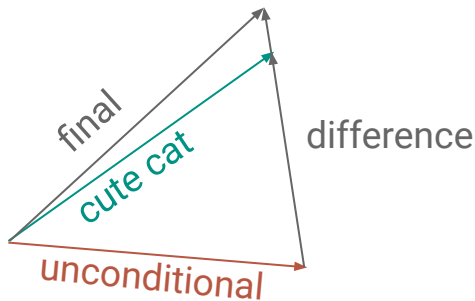
Classifier (Free) Guidance

Classifier Guidance

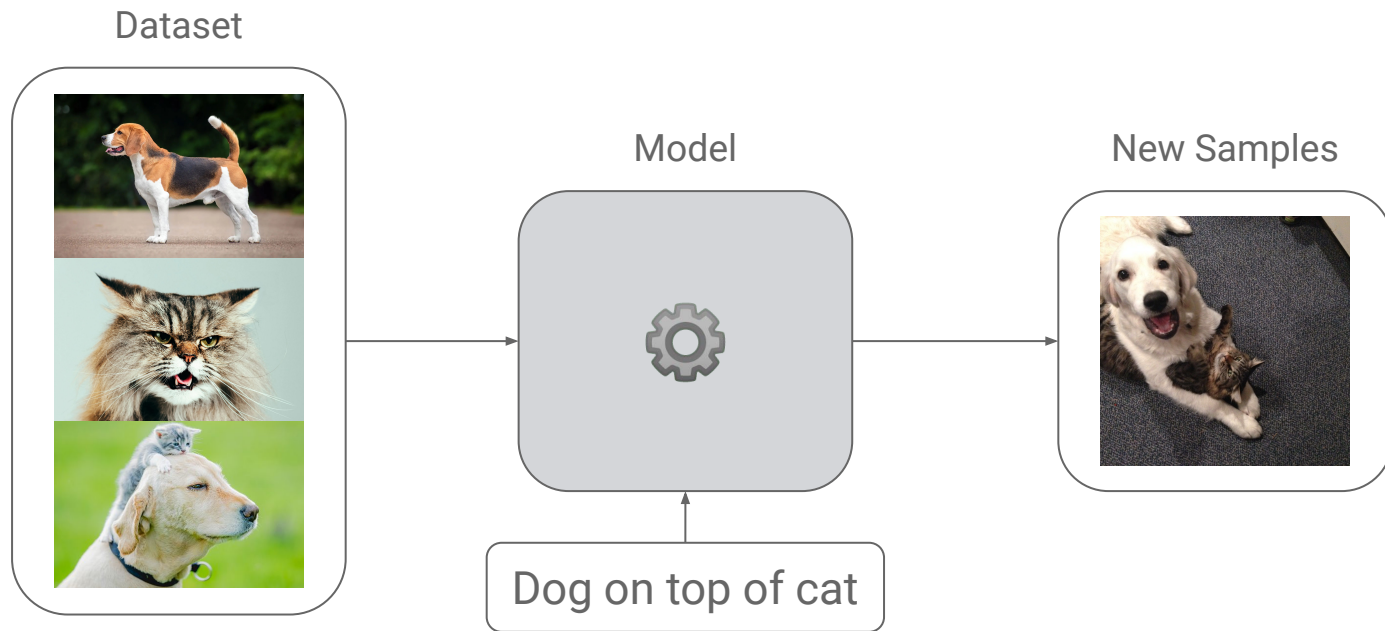
$$q_{\theta}(x_t, c) := q_{\theta}(x_t, c) - w\sigma_t \nabla_{x_t} \log(p_{\theta}(c|x_t))$$

Classifier Free Guidance

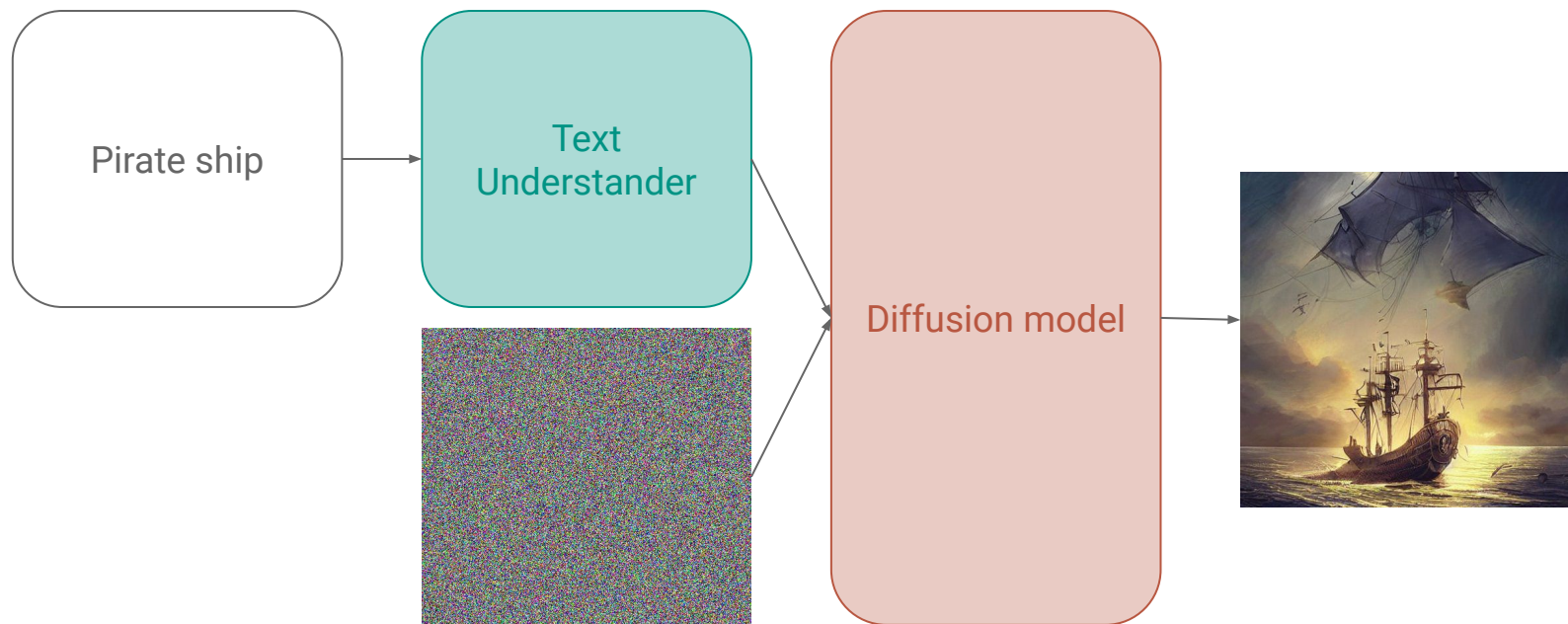
$$q_{\theta}(x_t, c) := (1 + w)q_{\theta}(x_t, c) - wq_{\theta}(x_t)$$



Task: custom prompts

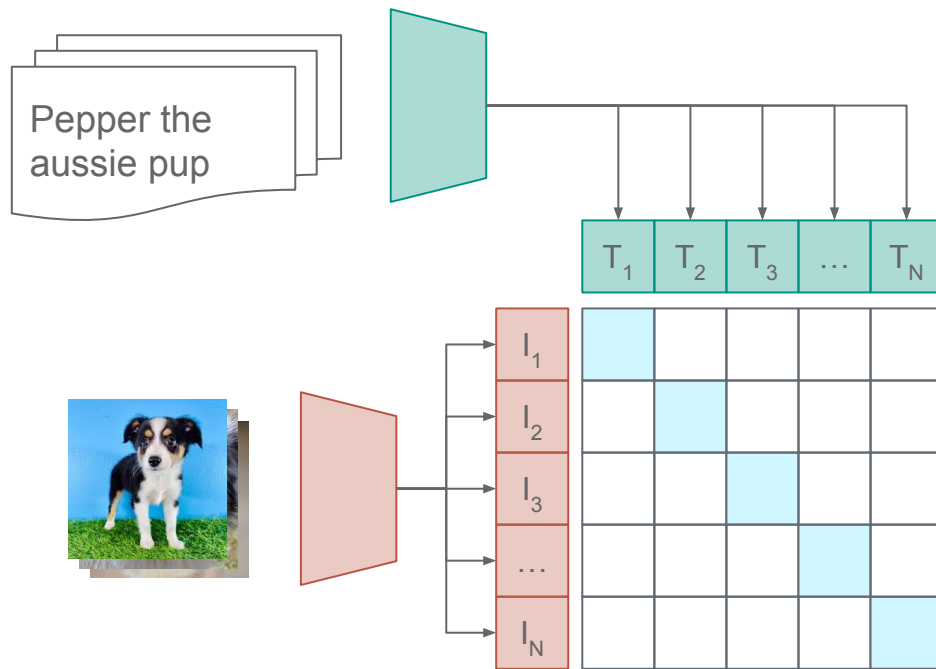


Stable Diffusion Architecture Overview



Relating text with images (CLIP)

Learning Transferable Visual Models
From Natural Language Supervision
Radford, Kim et al (Feb 2021)



Zero shot classification (CLIP)

Learning Transferable Visual Models
From Natural Language Supervision

Radford, Kim et al (Feb 2021)



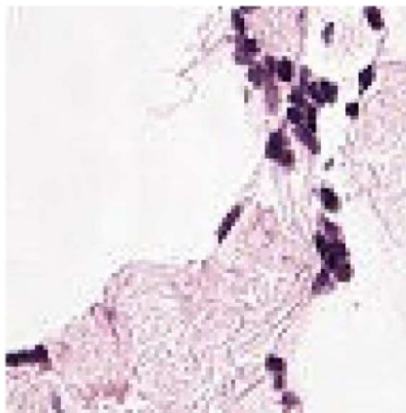
A photo of guacamole, a type of food

A photo of ceviche, a type of food

A photo of edamame, a type of food

A photo of tuna tartare, a type of food

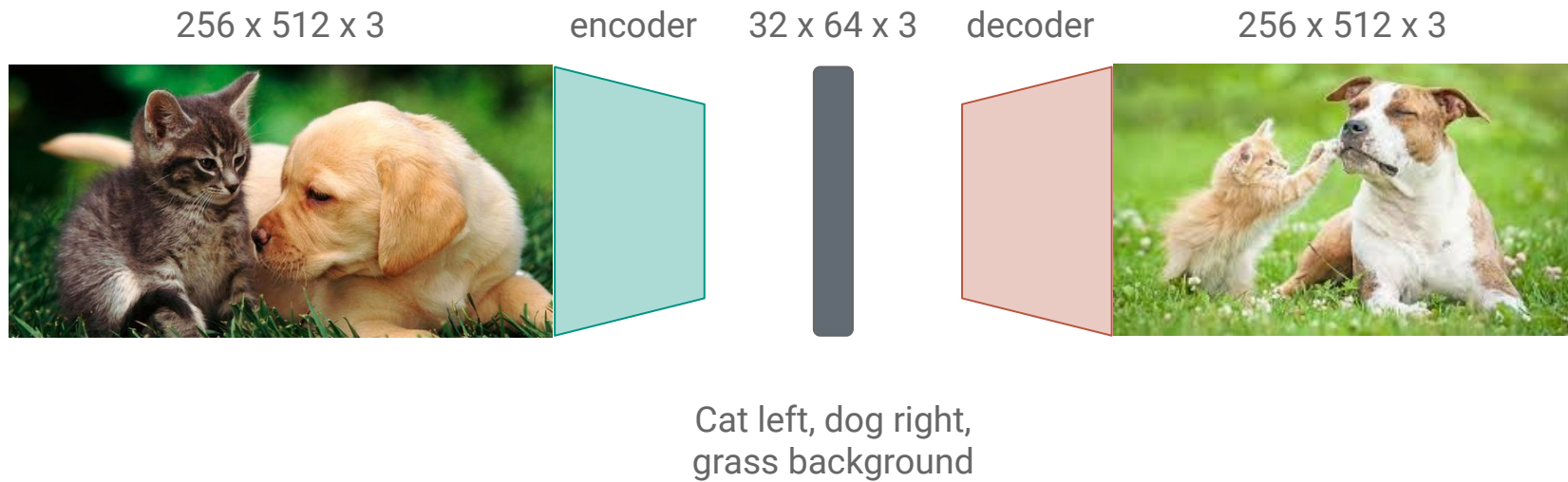
A photo of hummus, a type of food



this is a photo of lymph node tumor tissue

this is a photo of healthy lymph node tissue

Latent space



Stable Diffusion

High-Resolution Image Synthesis with
Latent Diffusion Models

Rombach, Blattmann et al. (Apr 2022)

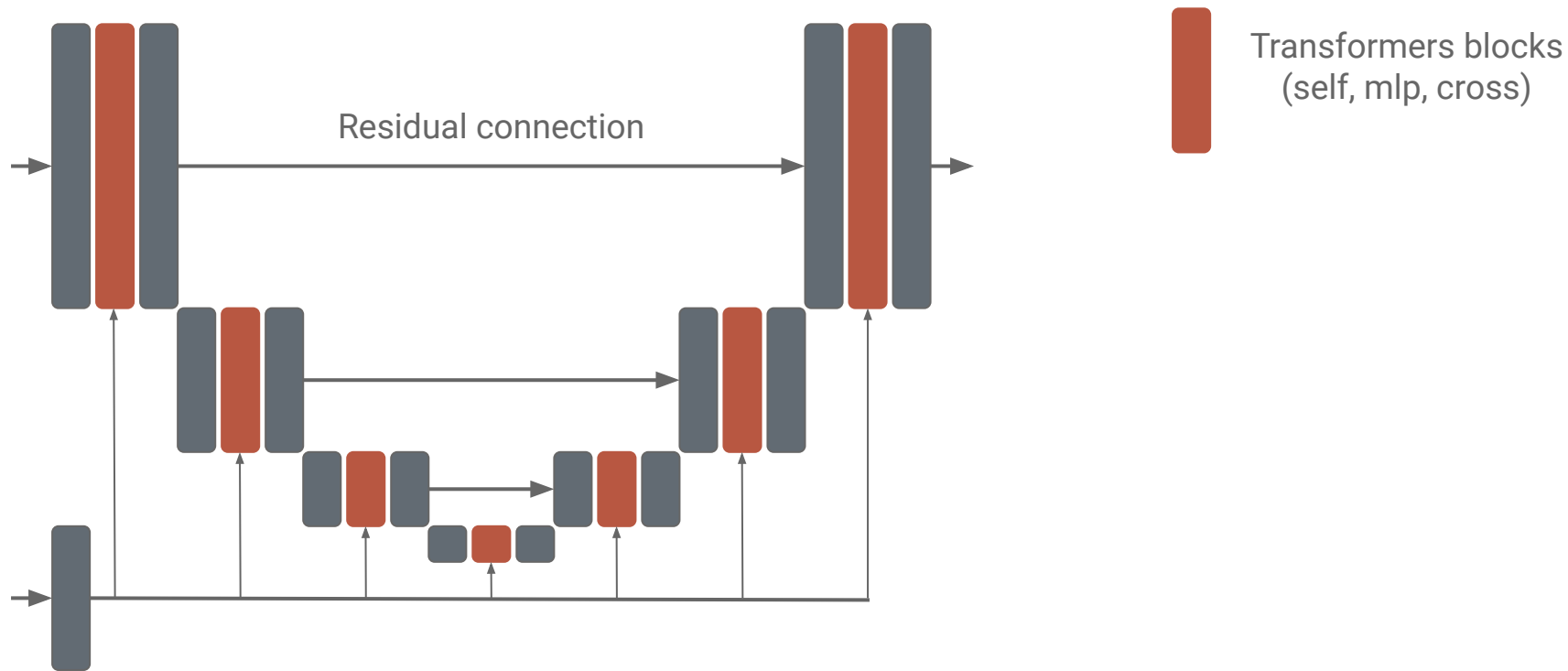
Latent diffusion models (LDM)

- Previous algorithms diffused in pixel space
- Diffusing in latent space is more efficient

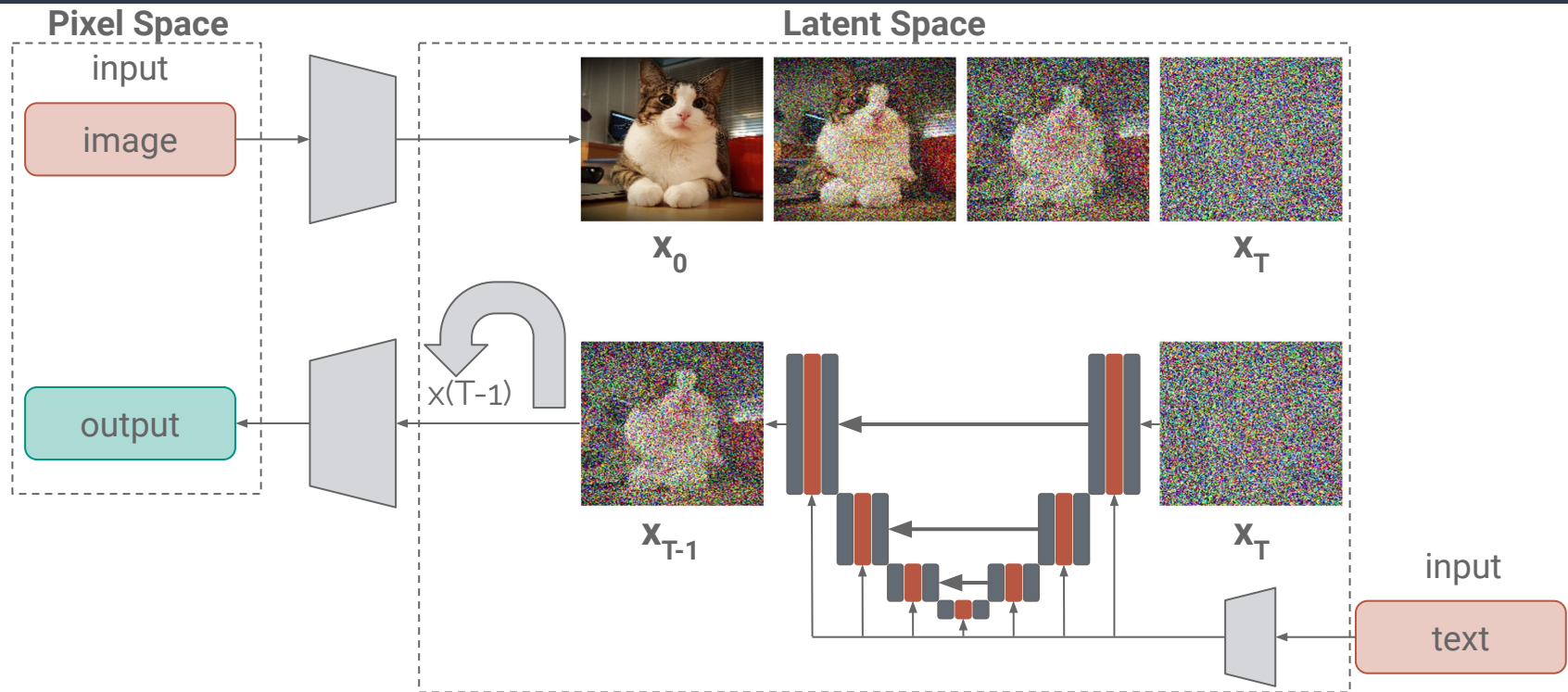
Incorporating text

- Introducing cross-attention with some input (text embeddings, reference images, ...)
- Full transformers decoder blocks in Unet
- Query from U-net, Key/Value from image/text

U-net with cross-attention



Overview

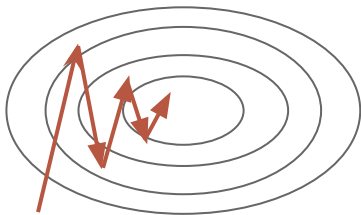


Rombach, Blattmann et al. (Apr 2022)

Samplers & Optimisation

Without momentum

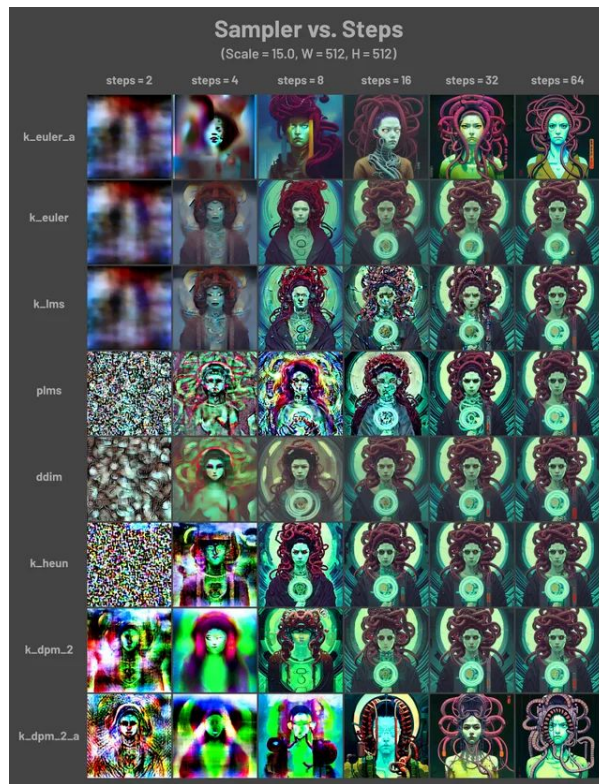
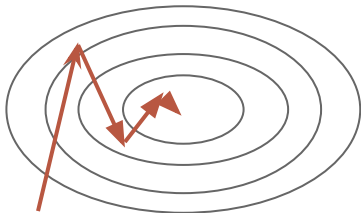
$$\theta := \theta - \alpha \nabla_{\theta} J(\theta)$$



With momentum

$$v_t = \beta v_{t-1} + (1 - \beta) \nabla_{\theta} J(\theta)$$

$$\theta := \theta - \alpha v_t$$



Dataset & Controversies

LAION-5B: An open large-scale dataset
for training next generation image-text
models

Schuhmann, Beaumont et al. (Oct 2022)

5 billion text image pairs from across the internet

- Text classifier to detect english (2.2b)
- Aesthetic filter with ($> 4.5/10$) (1.2b)
- NSFW filter (> 0.1) (1.1b)

Contains public-space artwork

The data can comfortably be downloaded with
img2dataset (240TB)

Stable diffusion v2

OpenCLIP instead of CLIP as encoder

- Less celebrity and artist knowledge
- More attention to negative prompts

More features

- Higher native resolution (512x512)
- More parameters
- Native upscaling model
- Depth2Img model

Images

Practical info about image generation

Images & prompts



kneeling cat knight,
portrait, finely detailed
armor, intricate design,
silver, silk, cinematic
lighting, 4k



full body shot, benedict
cumberbatch as a
(cucumber:1.1) knight

intricate, highly detailed,
digital painting, artstation,
concept art

art by artgerm and greg
rutkowski and alphonse
mucha, full body shot

**deformed, bad anatomy,
disfigured, gore**

Images & prompts



fantasy japanese spring forest, river, colorful, flowers, green grass, trees in background, greg rutkowski, masterpiece, intricate, trending on artstation



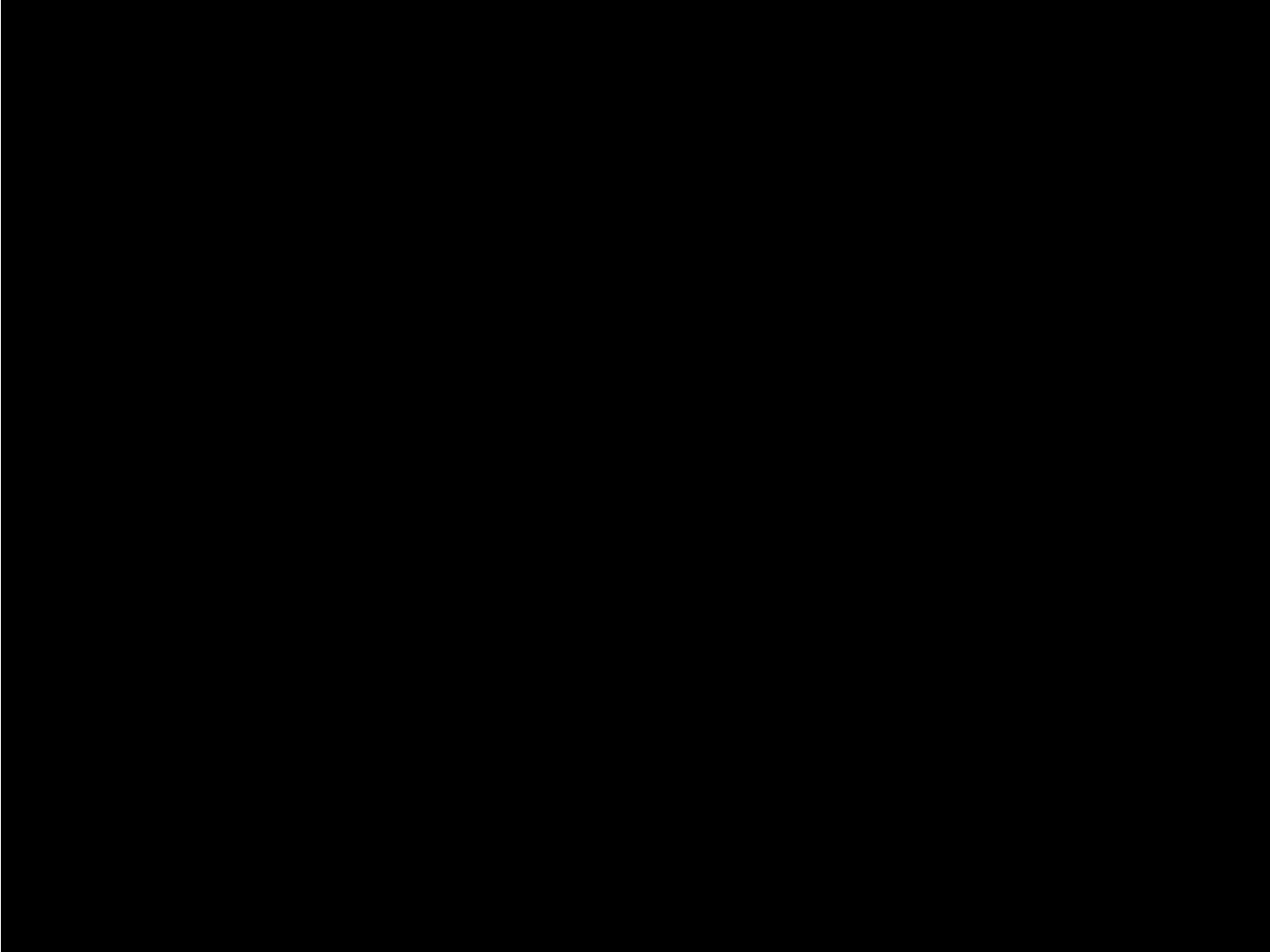
epic snowy mountains, dawn, rocks, realistic lighting, skyrim, greg rutkowski, masterpiece, intricate, trending on artstation

Quirks

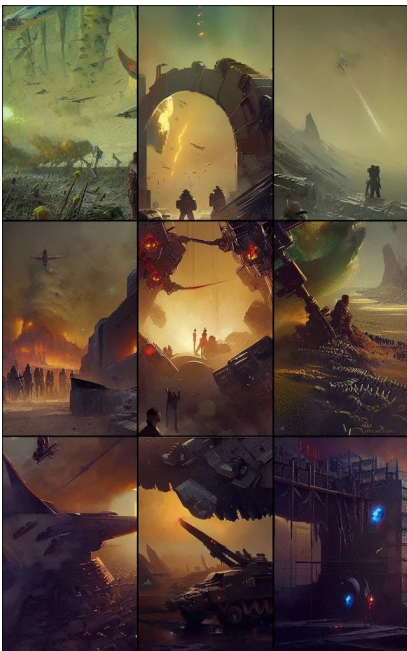
Everyone: AI art will make designers obsolete

AI accepting the job:





More cool stuff



How to draw an owl

1.



1. Draw some circles



2. img2img: "An owl perched on a branch, high-quality, Pixiv, WLOP, Greg Rutkowski, ArtStation. Oil on canvas."

Proprietary

How are the big labs tackling the problem?

DALL·E 2

Hierarchical Text-Conditional
Image Generation with CLIP Latent
Ramesh, Dhariwal et al. (Nov 2022)

OpenAI

Very similar to stable diffusion V2

Uses pixel space and not latent space

Better images due to high quality training data

Different datasets

Imagen

Photorealistic Text-to-Image Diffusion
Models with Deep Language
Understanding
Saharia, Chan (May 2022)

Google Brain

CLIP vs Large Language Models (LLM)

- CLIP is trained on image-text pairs
- T5 and BERT are trained on text only

Assumptions in previous papers

CLIP should have more useful embeddings for images

But actually ...

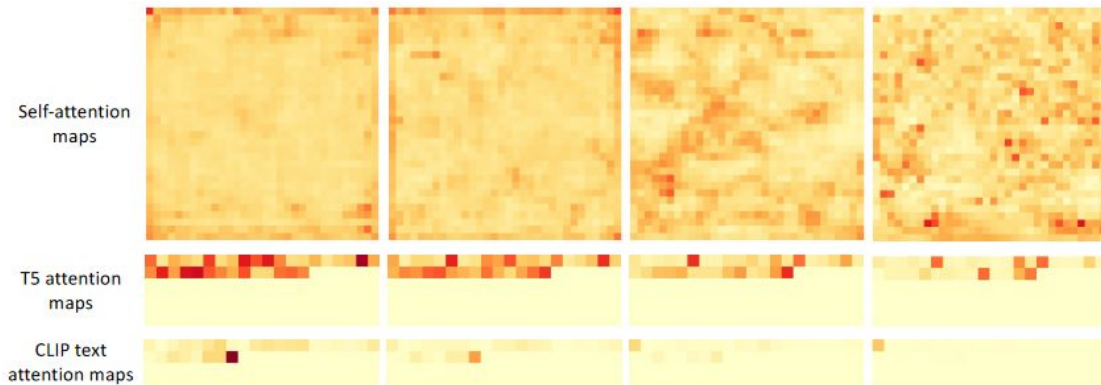
Scaling the LLM, not the diffuser, improves the images

eDiff-I

eDiff-I: Text-to-Image Diffusion Models
with an Ensemble of Expert Denoisers

Balaji, Nah et al. (Nov 2022)

NVIDIA



Single diffusion model for all noise levels isn't enough

- Cannot pick up on small details
- Doesn't scale well for training / performance

Proposed solutions

- Use a Mixture of Experts for varying noise levels
- Use both T5 and CLIP

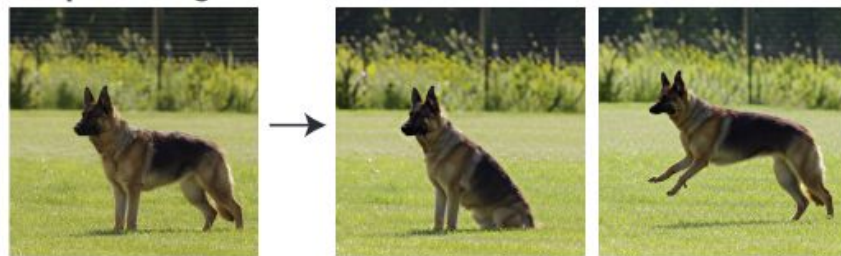
Future

This is the last chapter, I swear

Selective editing

Imagic: Text-Based Real Image Editing
with Diffusion Models

Kawar, Zada et al. (Nov 2022)

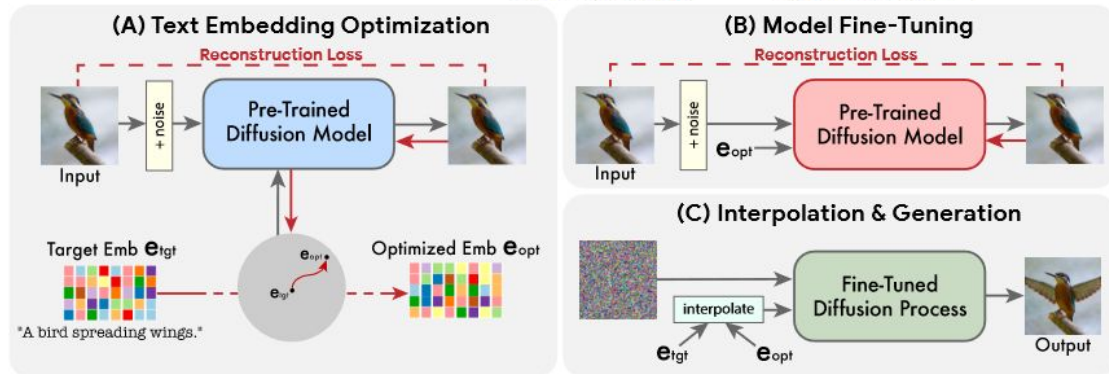


"A sitting dog"

"A jumping dog"

"A person giving
the thumbs up"

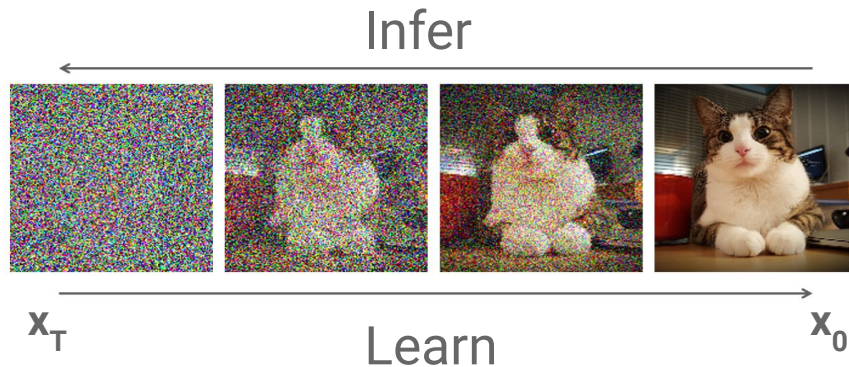
"A person with
crossed arms"



Reducing the amount of steps

On Distillation of Guided Diffusion Models

Meng et al. (Dec 2022)



Train a student model which distils the learnt model

The student is trained to perform several steps at a time

Future improvements

Three Challenges Ahead for Stable
Diffusion

Anderson (December, 2022)

Tile based pipelines

- Hardware limits for generating large images
- Would allow to focus attention precisely

Addressing issues with human limbs

- Humans are good at recognising 'bad' faces
- Objects require precise long-range coordination

Customization

- Better textual inversion
- Less expensive training

Questions